Certificate-Only Causal Discovery: Anytime-Valid Adjacency and Orientation

Quang-Vinh Dang[1*]

[1*]British University Vietnam, Hung Yen, Vietnam.

Corresponding author(s). E-mail(s): vinh.dq4@buv.edu.vn;

Abstract

Constraint-based causal discovery removes an edge when a conditional-independence test *fails to reject*, converting absence of evidence into evidence of absence. This is why finite-sample correctness for such methods requires faithfulness, and it is why their output cannot distinguish a feature the data support from one about which nothing was learned. We develop a discovery algorithm built on a single rule: assert a feature of the graph only by *rejecting* a null that is true whenever that feature is absent, and never assert anything by failing to reject. Two certificates implement the rule. For adjacency, the statement "some conditioning set separates the pair" is a union null, and the minimum of the per-set e-processes is an e-process for it; crossing a threshold certifies an edge under the Markov condition and a bound on separator size, given a correctly specified per-set test, with no faithfulness assumption. For orientation, the two directions of a pair induce different joint densities under a linear non-Gaussian model, so a sequential universal-inference e-process against the reversed factorisation certifies an arrowhead, and abstains automatically under Gaussian noise, where the direction is not identified. Both certificates are anytime-valid, so the estimate may be monitored continuously and the run stopped by any data-dependent rule. The resulting algorithm, CERT-CD, grows from the empty graph and reports undecided pairs as undecided. The adjacency certificate bounds by $\boldsymbol{\alpha}$, unconditionally, the probability that any certified edge is *ever* absent, uniformly over time; the same bound for a certified arrowhead holds only conditional on the linear non-Gaussian model actually describing the pair given its frozen conditioning block, a premise correlated with faithfulness through the order in which pairs are certified—not a contradiction of the theorem, which never claims the bound unconditionally, but exactly the gap a reader must know before trusting an arrowhead. On simulated instances that violate $\boldsymbol{\delta}$-strong faithfulness, the adjacency certificates stay within budget while methods that delete on non-rejection lose a true edge in most runs; the orientation certificates exceed theirs on those same instances, and we identify and measure the mechanism responsible. On premise-satisfying instances **93%** of true edges are certified and oriented, against a Markov-equivalence-class orientation ceiling of **33%** on the same denominator. On the observational protein-signalling data of Sachs and colleagues the two halves of the output diverge instructively: every certified edge lies in the published consensus skeleton, while every certified arrowhead is reversed relative to it—a disagreement we trace to the linear non-Gaussian model rather than to the inferential machinery, since DirectLiNGAM reverses the same arcs.

Keywords: causal discovery, e-values, anytime-valid inference, safe testing, family-wise error rate, LiNGAM

MSC Classification: 62H22 , 62L12 , 62F03

1 Introduction

A causal discovery algorithm returns a graph, and a practitioner reading that graph would like to know which of its features the data support. Causal graphical models have become a standard language for that question (Pearl, 2009; Peters, Janzing, & Schölkopf, 2017), and the algorithms that estimate them from observational data are correspondingly consequential. Methods in the constraint-based tradition—PC, FCI and their descendants (Colombo, Maathuis, Kalisch, & Richardson, 2012; Spirtes, Glymour, & Scheines, 2000; Spirtes, Meek, & Richardson, 1995)—answer this question poorly, and the reason is structural rather than incidental. They begin from the complete graph and delete an edge whenever a conditional-independence test fails to reject. Failure to reject is not evidence of independence. It is compatible with independence, with a dependence too weak for the test to see at the current sample size, and with a test that had little power against the alternative in question. The deletion happens regardless, and the resulting absence of an edge is reported with the same authority as the presence of one.

Two consequences follow, and both are well known in their separate literatures. First, finite-sample guarantees for these methods require faithfulness or a quantitative strengthening of it, because only such an assumption converts "the test did not reject" into "the variables are independent". Robins, Scheines, Spirtes, and Wasserman (2003) showed that no uniformly consistent estimator of causal structure exists without such a strengthening, and Uhler, Raskutti, Bühlmann, and Yu (2013) showed that the set of parameter values violating λ-strong faithfulness has substantial measure even in modest graphs. The assumption is doing work the data cannot do. Second, the output of these methods is two-valued where the evidence is three-valued: a missing edge means either "the data indicate no edge" or "nothing was established about this pair", and the graph does not say which.

This paper takes the opposite discipline. We adopt a single rule and follow it to its consequences:

> *Assert a feature of the graph only by rejecting a null hypothesis that is true whenever that feature is absent. Never assert anything by failing to reject.*

We call this the *certificate-only* rule, and an assertion licensed by such a rejection a *certificate*. The rule is restrictive, and its consequences are not cosmetic. An algorithm obeying it cannot begin from the complete graph, because it has no licence to delete; it must begin from the empty graph and add. It cannot report an edge as absent, only as undecided. And the quantity it controls is the probability of asserting something false, not the probability of missing something true. Power becomes a descriptive property of a run rather than part of the guarantee.

Making the rule operational requires two certificates, and their statistical situations are quite different.

Adjacency.

For a pair (i, j), the null that is true whenever the edge is absent is composite: *some* conditioning set separates them. This is a union over conditioning sets, and the minimum of the per-set e-processes is an e-process for a union null (Section 4). Two consequences matter more than the construction itself. Validity uses no faithfulness assumption—if i and j are non-adjacent in the true graph then $\mathrm{pa}(i)$ or $\mathrm{pa}(j)$ separates them, so the null holds and Ville's inequality applies—and the multiplicity correction runs over pairs rather than over (pair, conditioning set) triples, because one hypothesis is tested per pair.

Orientation.

Orientation error is not uncontrolled in the literature. PC-p (Strobl, Spirtes, & Visweswaran, 2019) attaches edge-specific p-values to unshielded colliders and Meek-rule orientations and controls their false discovery rate; Shimizu and Kano (2008) test the direction of causation between two variables using non-normality; Strieder and Drton (2023) construct confidence statements for causal effects under structure uncertainty. What has not been available is orientation error control that is simultaneously finite-sample, time-uniform, family-wise over a whole graph, and not confined to the Markov equivalence class. Constraint-based orientation tests are confined to that class by construction, so an edge left undirected in the CPDAG is unorientable in principle by any of them. The functional-model methods that escape the class—LiNGAM and its descendants (Hyvärinen & Smith, 2013; Shimizu, Hoyer, Hyvärinen, & Kerminen, 2006; Shimizu et al., 2011)—choose a direction by comparing scores, and the comparison itself carries no error statement. Under a linear non-Gaussian model the two orientations of a pair induce different joint densities and exactly one is correct, so "the direction is $j \to i$" is a genuine null hypothesis; a sequential universal-inference e-process against it certifies the arrowhead $i \to j$ (Section 5). Under Gaussian noise the two factorisations are observationally indistinguishable, both directions' nulls hold simultaneously, and neither is certified with probability more than α—the procedure abstains exactly where the direction is unidentified, at the same level that governs every other certificate, without being told to.

Why anytime-valid.

Both certificates are built as e-processes rather than as fixed-sample tests, and the choice is not decorative. A discovery run on streaming or accumulating data invites the analyst to look at the current graph, and looking is a form of optional stopping. A fixed-sample guarantee is void the moment the analyst stops on the basis of what they saw. Section 9.2 quantifies the gap: at nominal $\alpha = 0.05$, a Fisher-z test inspected after every observation rejects a true null in 34.6% of runs, against 5.6% when it is evaluated once. An e-process pays for time-uniformity in power, and we report that price rather than eliding it.

Contributions.

The paper makes four.

1. An e-process for the composite *separability* null, obtained as a minimum over conditioning sets, giving time-uniform certification of adjacency under the Markov condition and a bound on separator size, with no faithfulness assumption (Section 4). We give the exact per-set primitive for the linear-Gaussian case in closed form, together with the confidence sequence obtained by inverting it.
2. A sequential certificate for causal *orientation* (Section 5) that is valid at any stopping time, is not confined to the Markov equivalence class, and abstains automatically under Gaussianity. We identify a requirement the construction imposes on the fitting procedure—the null-model fit must genuinely attain its supremum—which is easy to violate and which silently inflates the e-process when violated (Section 5.3).
3. CERT-CD, an algorithm combining both certificates under one budget (Section 6). Its output is a three-valued graph in which every asserted edge and arrowhead carries a certificate and undecided pairs are explicit, with a whole-graph guarantee that is proved rather than assumed.
4. An empirical study (Section 9) that stratifies instances by whether the premise of the competing guarantee holds, rather than pooling them; measures the assumptions of our own theorem instead of asserting them; reports the regimes in which the certificates are void (Section 9.12); and applies the method to the observational protein-signalling data of Sachs, Perez, Pe'er, Lauffenburger, and Nolan (2005).

The study is not uniformly favourable to the method, and the two places where it is not are the more informative ones. On instances violating δ-strong faithfulness the orientation certificates exceed their family-wise budget (Section 9.5), because the conditioning block they freeze is built from already-certified neighbours and is therefore incomplete exactly when certification is slow. On the protein-signalling data every certified arrowhead is reversed relative to the published consensus (Section 9.10), and DirectLiNGAM reverses the same arcs, which locates the failure in the shared functional model rather than in the inference. Both are instances of one gap—the direction null assumes a model, and a certificate is only as good as its null—and we have tried to give them the prominence a guarantee-bearing method owes them.

We are explicit throughout about which components are standard. Section 8 gives that accounting in detail, including three claims of novelty we made in earlier drafts and then withdrew after reading the prior art.

Notation and outline.
Section 2 fixes notation and reviews e-processes and the graphical setting. Section 3 states the certificate-only rule formally and derives its structural consequences. Sections 4 and 5 develop the two certificates. Section 6 assembles CERT-CD and proves the whole-graph guarantee; Section 7 extends it to latent confounders. Section 8 positions the work. Section 9 reports experiments and Section 10 discusses limitations. Proofs and derivations omitted from the main text are in the appendices.

2 Background

2.1 E-variables, e-processes and Ville's inequality

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space and let H_0 be a set of distributions on it.

Definition 1 (E-variable) A non-negative random variable E is an *e-variable* for H_0 if $\mathbb{E}_P[E] \leq 1$ for every $P \in H_0$.

Definition 2 (E-process) A non-negative, adapted process $(E_t)_{t\geq 0}$ is an *e-process* for H_0 if $\mathbb{E}_P[E_\tau] \leq 1$ for every $P \in H_0$ and every stopping time τ (finite or not, with $E_\infty := \liminf_t E_t$).

Every non-negative supermartingale starting at one is an e-process, by optional stopping (Shafer & Vovk, 2019); the converse fails, and the distinction matters for composite nulls, where an e-process need not be a supermartingale under any single null distribution (Ramdas, Grünwald, Vovk, & Shafer, 2023). Both of our constructions are e-processes that are dominated pointwise by a P-supermartingale for each $P \in H_0$, with the dominating process depending on P. That is the standard route to an e-process for a composite null, and it is the route both proofs below take.

The operational content of Definition 2 is Ville's inequality (Ville, 1939): for an e-process (E_t) and any $\alpha \in (0, 1)$,

$$\mathbb{P}\big(\exists\, t \geq 0 : E_t \geq 1/\alpha\big) \;\leq\; \alpha \qquad \text{for every } P \in H_0. \tag{1}$$

The quantifier is inside the probability. This is what licenses continuous monitoring: an analyst may inspect the evidence after every observation and stop whenever they wish, and the probability that the process *ever* crosses the threshold under the null is still at most α. A p-value offers no analogue. A fixed-sample p-value controls the rejection probability at one pre-specified sample size, and the guarantee is destroyed by looking; the phenomenon is familiar from sequential clinical trials and from online experimentation (Johari, Koomen, Pekelis, & Walsh, 2022).

E-values also compose in ways p-values do not, and we use two such composition rules. If $E^{(1)}, \ldots, E^{(m)}$ are e-variables for $H_0^{(1)}, \ldots, H_0^{(m)}$ then $\min_k E^{(k)}$ is an e-variable for $\bigcup_k H_0^{(k)}$, and any convex combination $\sum_k w_k E^{(k)}$ with $w_k \geq 0$, $\sum_k w_k = 1$

is an e-variable for $\bigcap_k H_0^{(k)}$, under *arbitrary* dependence between the components (Vovk & Wang, 2021). The first rule is what makes the adjacency certificate possible; the second is what makes the multiplicity discussion of Section 6.4 non-trivial.

Recent work has developed e-processes for a wide range of settings—bounded means (Waudby-Smith & Ramdas, 2024), linear models (Lindon, Ham, Tingley, & Bojinov, 2026), stratified counts (Turner & Grünwald, 2023)—and has characterised admissible constructions (Ramdas et al., 2023). Two general recipes recur below. The first is the *method of mixtures*: under a group-invariant null, integrating the likelihood ratio against the right-Haar measure on the nuisance group yields an exact test martingale (Berger, Pericchi, & Varshavsky, 1998; Grünwald, de Heide, & Koolen, 2024; Hendriksen, de Heide, & Grünwald, 2021). Section 4.3 instantiates it for partial correlation. The second is *universal inference* (Wasserman, Ramdas, & Balakrishnan, 2020): for a null model $\{D_\theta : \theta \in \Theta_0\}$ and any alternative fitted on past data, the ratio of a sequentially-fitted alternative likelihood to the maximised null likelihood is an e-process, with no regularity conditions on Θ_0 beyond the existence of the supremum. Section 5 instantiates it for causal direction. Universal inference is known to be conservative (Tse & Davison, 2022), and Section 9.7 measures how conservative it is here.

2.2 Causal discovery setting

Let $\mathcal{G}^\star$ be a directed acyclic graph (DAG) with vertex set $V = \{1, \dots, d\}$ identified with random variables $X_1, \dots, X_d$, and let P be the joint law of $X = (X_1, \dots, X_d)$. We write $\mathrm{pa}(i)$ for the parents of i in $\mathcal{G}^\star$ and $\mathrm{de}(i)$ for its descendants. Throughout, observations $o_1, o_2, \dots$ are drawn i.i.d. from P and arrive in batches; $\mathcal{F}_t$ is the σ-field generated by the first t observations.

Assumption 1 (Markov condition) P is Markov to $\mathcal{G}^\star$: whenever S d-separates i from j in $\mathcal{G}^\star$, $X_i \perp\!\!\!\perp X_j \mid X_S$ under P.

Assumption 1 is the direction of the correspondence that follows from the causal structure itself: it says that the graph's separations are reflected in the distribution. Its converse—faithfulness, that every conditional independence in P corresponds to a d-separation in $\mathcal{G}^\star$—does not follow from the causal structure, and is the assumption we do not make for validity.

Write $\mathcal{S}_k(i, j) = \{S \subseteq V \setminus \{i, j\} : |S| \le k\}$ for the family of admissible conditioning sets. This family is determined by d and k and is fixed before any data are seen; nothing about it adapts to the sample. We say the pair (i, j) is *separable within* $\mathcal{S}_k$ under P if $X_i \perp\!\!\!\perp X_j \mid X_S$ for some $S \in \mathcal{S}_k(i, j)$.

Assumption 2 (Separator size) Every non-adjacent pair of $\mathcal{G}^\star$ has a d-separating set of size at most k.

Assumption 2 is structural rather than distributional: it constrains $\mathcal{G}^\star$, not P, and it can be guaranteed by choosing k large enough. It holds whenever k is at least the maximum in-degree of $\mathcal{G}^\star$, since for non-adjacent i, j either $j \notin \mathrm{de}(i)$, in which case $\mathrm{pa}(i)$ d-separates them, or $i \notin \mathrm{de}(j)$, in which case $\mathrm{pa}(j)$ does. Taking $k = d-2$ makes it vacuous at the cost of an exponential family $\mathcal{S}_k$; the trade-off is computational, not inferential, and we return to it in Section 6.3. Section 9.4 measures how often Assumption 2 holds at the k we use, and compares it with how often δ-strong faithfulness holds on the same instances.

2.3 Faithfulness and its quantitative strengthenings

Since the argument of this paper is about what faithfulness is doing, it is worth being precise about the versions in play.

Faithfulness holds if every conditional independence in P is implied by d-separation in $\mathcal{G}^\star$; Meek (1995b) showed that it fails only on a Lebesgue-null set of parameters for a fixed graph, which is the result usually cited in its defence and which Uhler et al. (2013) showed to be misleading in finite samples. *Adjacency-faithfulness* (Ramsey, Spirtes, & Zhang, 2006) is the weaker requirement that adjacent variables are dependent given every conditioning set. *δ-strong faithfulness* (Kalisch & Bühlmann, 2007; Uhler et al., 2013) is the quantitative strengthening that every partial correlation not forced to zero by d-separation has absolute value at least δ; it is what turns consistency into a finite-sample statement, and it is the premise of the comparators in Section 9.5.

The asymmetry we exploit is this. Faithfulness in any of its forms is an assumption about which *dependences* exist. A procedure that asserts an edge by rejecting an independence null needs it only for *power*: without it, a genuinely adjacent pair whose dependence is cancelled will fail to be certified, and the pair stays undecided. A procedure that asserts a *non*-edge by failing to reject needs it for *validity*: without it, a cancelled dependence produces a confidently deleted edge. The same assumption sits on opposite sides of the guarantee depending on which direction of the inference the algorithm runs. J. Zhang and Spirtes (2008) and Ramsey et al. (2006) respond to this by detecting or tolerating unfaithfulness within a delete-on-non-rejection architecture; we respond by changing the architecture.

3 The certificate-only principle

We state the rule formally and record what it forces.

Definition 3 (Certificate) Let ϕ be a feature of $\mathcal{G}^\star$—a proposition such as "i and j are adjacent" or "$\mathcal{G}^\star$ contains the arrowhead $i \to j$". A *certifying null* for ϕ is a hypothesis H_ϕ with the property that $P \in H_\phi$ whenever ϕ is false. A *certificate for ϕ at level α* is the event that an e-process for H_ϕ exceeds $1/\alpha$.

The implication in Definition 3 runs one way only, and that is the whole point. H_ϕ must be true whenever ϕ is false; it may also be true in some cases where ϕ is true, and then the certificate simply never fires. This asymmetry is what buys validity without faithfulness, and what costs power.

Proposition 1 (Soundness of a single certificate) *If (E_t) is an e-process for a certifying null H_ϕ, then $\mathbb{P}(\exists t : \phi$ is certified at level α and ϕ is false$) \leq \alpha$.*

Proof If ϕ is false then $P \in H_\phi$ by Definition 3, so (E_t) is an e-process under P and (1) bounds the probability that it ever exceeds $1/\alpha$ by α. If ϕ is true the event in question is empty. □

Proposition 1 is elementary; we state it to make explicit that the burden of the design is entirely in constructing H_ϕ, not in the inference. Three structural consequences follow, and they shape everything below.

The algorithm cannot delete.

There is no certifying null for the proposition "i and j are non-adjacent". Such a null would have to be true whenever the pair *is* adjacent, that is, it would have to be a dependence hypothesis rejected by evidence of independence; but evidence of independence is exactly what a hypothesis test does not provide. Equivalence testing supplies the nearest available substitute—one can reject "the partial correlation exceeds δ in absolute value"—but the resulting statement is "the association is smaller than δ", which is a claim about effect size, not about absence. We return to this in Section 6.5. In the absence of such a null, an algorithm obeying the rule must start from the empty graph.

The output is three-valued.

Each pair is in exactly one of three states: *certified adjacent* (with or without a certified orientation), *undecided*, or—if the extension of Section 6.5 is enabled—*certified negligible at margin δ*. The absence of an edge in the output graph is not an assertion.

Power is descriptive, not guaranteed.

Nothing in Proposition 1 promises that a true edge is ever certified. Whether it is depends on adjacency-faithfulness, on the sample size, and on the primitive's power. This is a real limitation and we treat it as one: Section 9 reports certification rates alongside error rates throughout, and reports the fraction of pairs left undecided rather than suppressing it.

4 The adjacency certificate

4.1 The separability null

Fix a pair (i, j) and consider

$$H_{ij}^{\text{sep}} \;:\quad \exists\, S \in \mathcal{S}_k(i,j) \;\text{ such that }\; X_i \perp\!\!\!\perp X_j \mid X_S. \tag{2}$$

Lemma 2 *Under Assumptions 1 and 2, H_{ij}^{sep} is a certifying null for the adjacency of i and j.*

Proof Suppose i and j are non-adjacent in $\mathcal{G}^\star$. By Assumption 2 there is a set S with $|S| \le k$ that d-separates them, and $S \subseteq V \setminus \{i, j\}$, so $S \in \mathcal{S}_k(i, j)$. By Assumption 1, $X_i \perp\!\!\!\perp X_j \mid X_S$ under P. Hence $P \in H_{ij}^{\text{sep}}$. □

Note what the lemma does *not* require. It says nothing about adjacent pairs. If i and j are adjacent, P may or may not lie in H_{ij}^{sep}: an adjacent pair whose dependence is exactly cancelled by a path through some S satisfies the null too, and then the certificate cannot fire. That is the failure of adjacency-faithfulness, and it costs power, not validity.

4.2 The minimum construction

Let $(E_t^{(S)})_{t\ge 0}$ be an e-process for the point null $X_i \perp\!\!\!\perp X_j \mid X_S$, for each $S \in \mathcal{S}_k(i, j)$, all built from the same data stream. Define

$$A_t(i, j) \;=\; \min_{S\in\mathcal{S}_k(i,j)} E_t^{(S)}(i, j). \tag{3}$$

Proposition 3 *$(A_t)_{t\ge 0}$ is an e-process for H_{ij}^{sep}.*

Proof Let $P \in H_{ij}^{\text{sep}}$. By (2) there exists $S^\star \in \mathcal{S}_k(i, j)$ with $X_i \perp\!\!\!\perp X_j \mid X_{S^\star}$ under P, so P satisfies the point null for which $(E_t^{(S^\star)})$ is an e-process. By (3), $A_t \le E_t^{(S^\star)}$ pointwise for every t, hence $A_\tau \le E_\tau^{(S^\star)}$ for every stopping time τ, and non-negativity gives $\mathbb{E}_P[A_\tau] \le \mathbb{E}_P[E_\tau^{(S^\star)}] \le 1$. □

Combining Lemma 2, Proposition 3 and Proposition 1 gives the adjacency guarantee: the probability that $A_t(i, j)$ ever exceeds $1/\alpha$ for a non-adjacent pair is at most α.

That a minimum of e-values is an e-value for a union null is standard (Vovk & Wang, 2021), and the proof above is the standard one. What is specific here is the identification of adjacency as the complement of such a union, and the two consequences that follow.

Remark 1 (No faithfulness) The proof uses only that (2) is true for non-adjacent pairs, which follows from Assumptions 1 and 2. Faithfulness enters only through power: adjacency-faithfulness (Ramsey et al., 2006) is what makes every $E^{(S)}$ grow for a genuinely adjacent pair, and hence what makes the minimum grow. An instance where it fails leaves pairs undecided rather than producing wrong edges. Stated exactly, validity requires three things: the Markov condition, Assumption 2, and a valid per-set e-process. The third is not free. Our default primitive (Section 4.3) is a linear-Gaussian safe test, so that model is assumed *there*; substituting a nonparametric sequential test (He & Sutherland, 2026) removes it at a cost in power. Proposition 3 is indifferent to which primitive is used, and Section 9.3 measures the primitive's calibration under six noise laws, including three outside its model.

Remark 2 (Multiplicity over pairs) The construction tests one hypothesis per pair, not one per (pair, conditioning set) triple. A union bound over the adjacency family therefore runs over $\binom{d}{2}$ hypotheses rather than $\binom{d}{2} \cdot |\mathcal{S}_k|$. At $d = 6$ and $k = 2$ we have $|\mathcal{S}_k| = \binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 11$, so the per-hypothesis level is 11 times larger than a triple-indexed budget would allow. The saving grows with k: at $d = 10$, $k = 3$ it is 93. This is not a free lunch—the minimum in (3) is itself conservative, since it discards all but the least favourable conditioning set—but the conservatism is of a different kind from a multiplicity correction, and it is the kind that vanishes as the least favourable set becomes uninformative.

4.3 The per-set primitive

Proposition 3 needs an e-process for each point null $X_i \perp\!\!\!\perp X_j \mid X_S$. For linear-Gaussian data we use an exact one, obtained as a Bayes factor with right-Haar priors on the nuisance parameters. Write the hypothesis in regression form,

$$X_i \;=\; X_S^\top \alpha \;+\; \beta X_j \;+\; \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2), \tag{4}$$

so that the null is $\beta = 0$ with nuisance parameters (α, σ) shared between the hypotheses. Place the right-Haar prior $\pi(\alpha, \sigma) \propto \sigma^{-1}$ on the nuisance block under both hypotheses, and $\beta \mid \sigma \sim N(0, \sigma^2 g)$ on the tested coefficient under the alternative, in the manner of Zellner's g-prior (Zellner, 1986). The right-Haar choice on the nuisance block is the invariance-based prior of Jeffreys (1946) specialised to this group. Integrating both marginal likelihoods in closed form (Appendix A) gives, after n observations,

$$E_n \;=\; (1+G)^{-1/2} \left(1 - \kappa\, \hat{r}_n^2\right)^{-(n-p)/2}, \qquad G \;=\; g\, S_{yy}, \quad \kappa \;=\; \frac{G}{1+G}, \tag{5}$$

where $p = |S|+1$; S_{xx}, S_{xy}, S_{yy} are the cross-products of the X_S-residualised versions of X_j and X_i; and $\hat{r}_n^2 = S_{xy}^2/(S_{xx}S_{yy})$ is the squared sample partial correlation.

Proposition 4 *Under the linear-Gaussian model (4) with $\beta = 0$, the process (E_n) of (5) is a test martingale with respect to the filtration generated by the observations, and hence an e-process.*

The proof (Appendix A) is the standard right-Haar argument (Berger et al., 1998; Hendriksen et al., 2021): the null model is invariant under the group of location shifts in the X_S directions and scale changes, the right-Haar prior is relatively invariant under that group, and the resulting Bayes factor is a ratio of marginal likelihoods of the maximal invariant, whose null distribution is free of the nuisance parameters. Bayes factors of this form are exactly the ones for which optional stopping is valid, which is why not every Bayes factor would do here.

Two properties of (5) drive the implementation.

Sufficiency.

E_n depends on the data only through Schur complements of the running Gram matrix $\sum_{s \le n}(o_s - \bar{o})(o_s - \bar{o})^\top$. A single $O(d^2)$ sufficient statistic therefore serves the *entire* hypothesis family $\{(i,j,S)\}$ simultaneously: no per-hypothesis state is stored, and the cost of an update is $O(d^2)$ regardless of how many hypotheses are live. This is what makes it feasible to keep every pair's minimum in (3) current after every batch.

The prior scale must be fixed in advance.

g enters through $G = gS_{yy}$, and S_{yy} grows with n. A prior scale chosen to adapt to the running sample size—for instance by rescaling g so that G stays constant—destroys the martingale property, because the "prior" would then depend on the data it is being used to weigh. We fix g from a warm-up prefix of the stream that is excluded from every e-process, using it only to set standardisation constants. The warm-up length appears in the experiments as a tuning parameter and nowhere in the guarantee.

Remark 3 (Calibration was verified, not assumed) Proposition 4 is a theorem, but the implementation of a theorem can be wrong, and a miscalibrated primitive would invalidate every downstream claim while leaving all the proofs intact. We therefore verified calibration by simulation, reproducible by `experiments/exp1_type1_calibration.py --reps 3000 --nmax 1500`: monitoring (5) at a grid of sample sizes over 3000 replications up to $n = 1500$ under a true null gave realised crossing rates of 0.108, 0.051, 0.025 and 0.006 against nominal $\alpha \in \{0.20, 0.10, 0.05, 0.01\}$—in every case comfortably inside the nominal level, as Proposition 4 requires. Section 9.3 repeats the exercise under six noise laws.

4.4 Confidence sequence by inversion

Because (5) is available in closed form for every value of the tested coefficient, it can be inverted to give a confidence sequence—a sequence of intervals covering β simultaneously at all times with probability $1-\alpha$ (Howard, Ramdas, McAuliffe, & Sekhon, 2021). Writing $\hat{\beta} = S_{xy}/S_{yy}$ for the least-squares estimate of the coefficient of X_j in the residualised regression, the level-$(1-\alpha)$ confidence sequence is

$$\hat{\beta} \pm \frac{1}{S_{yy}}\sqrt{\frac{\tau\left(S_{xx}S_{yy} - S_{xy}^2\right)}{1-\tau}}, \qquad \tau = \frac{1}{\kappa}\left[1 - \left(\frac{\alpha}{\sqrt{1+G}}\right)^{2/(n-p)}\right], \tag{6}$$

valid whenever $\tau > 0$, and the whole line otherwise. The derivation is in Appendix B. We report it because it is the natural summary of evidence about an edge's strength, because it is what the equivalence-testing extension of Section 6.5 uses, and because it provides an independent check on the implementation: (6) must agree with a brute-force inversion of (5), and an early version of our code did not. The discrepancy was a sign error in the $\log(1+G)$ term, which produced intervals roughly half the correct width—a defect invisible in any Type-I error simulation of the test itself, since the test was correct and only the reported interval was wrong. Agreement between the closed form and a numerical inversion is now a unit test.

4.5 Beyond the linear-Gaussian primitive

Conditional independence testing is hard in general: Shah and Peters (2020) show that no test can have non-trivial power against all alternatives while controlling size over all null distributions with continuous conditional laws. Every usable primitive therefore buys power with a model. The two mainstream routes are the Model-X framework (Candès, Fan, Janson, & Lv, 2018), which assumes the conditional law of X_i given X_S is known, and parametric or kernel-based assumptions on the joint law. Sequential conditional-independence tests to date have largely taken the Model-X route; Csillag, Struchiner, and Goedert (2025) note that the requirement is typically inaccessible in a discovery setting, where the conditional laws are precisely what is unknown. Construction (5) avoids it by assuming linearity and Gaussianity instead, which is a strong assumption but a checkable one. Section 9.12 reports what happens when it fails.

5 The direction certificate

5.1 Why a direction can be a null hypothesis

Within a Markov equivalence class no observational statistic distinguishes the member DAGs, so "the direction is $j \to i$" is not a testable proposition there: both directions have the same likelihood under every distribution. The statement becomes testable exactly when a functional restriction breaks the symmetry, and the classical such restriction is linear non-Gaussianity.

Proposition 5 (Shimizu et al., 2006, after Darmois, 1953 and Comon, 1994) *Let $X_i = \beta X_j + e_i$ with $X_j \perp\!\!\!\perp e_i$, $\mathbb{E}[e_i] = 0$ and $\beta \neq 0$. If at most one of X_j, e_i is Gaussian, there is no δ, e_j with $X_j = \delta X_i + e_j$ and $X_i \perp\!\!\!\perp e_j$. If both are Gaussian, such a representation always exists.*

Proposition 5 is the identifiability statement that LiNGAM rests on, and it says two things we use in opposite ways. Under non-Gaussianity, exactly one of the two factorisations of the pair is compatible with the model, so the wrong one is a genuine, false, testable hypothesis. Under Gaussianity, both are compatible, so no evidence can separate them and any method that reports a direction anyway is reporting an artefact of its tie-breaking rule. Our construction inherits both halves.

The certifying null of Section 5.2 is conditional on a covariate block Z, and Proposition 5 is bivariate and unconditional; an earlier version of this paper cited the marginal non-Gaussianity of X_j and e_i as the licence for the conditional construction of Eqs. (7)–(8), and that citation does not bridge the gap on its own. The bridge is short, and we record it as a lemma because both Proposition 7 and Proposition 8 depend on it.

Lemma 6 (Conditional identifiability) *Suppose (X_i, X_j) given Z follows the $j \to i$ factorisation $D_\theta^{j\to i}$ of (7), with $u \perp\!\!\!\perp Z$ and $e \perp\!\!\!\perp (X_j, Z)$, and write $\tilde{e} = \beta u + e$ for the population*

Z-residual of X_i. If at most one of u, e is Gaussian, then (X_i, X_j) given Z does not also admit the reversed factorisation $D_\theta^{i\to j}$ of (8) for any θ. If both are Gaussian, it admits both.

Proof Because $e \perp\!\!\!\perp (X_j, Z)$ and u is $\sigma(X_j, Z)$-measurable, $u \perp\!\!\!\perp e$, and $\tilde{e} = \beta u + e$ is exactly the bivariate model of Proposition 5 with u in the role of its "X_j" and e in the role of its "e_i". If (X_i, X_j) given Z also admitted $D_\theta^{i\to j}$ for some $\theta = (b, \delta, \eta, \cdot)$, the same computation applied to (8) gives that the Z-residual of X_i is $v := X_i - b^\top Z$ and the Z-residual of X_j is $\delta v + w$, with $v \perp\!\!\!\perp w$. Population linear regression residuals on Z are unique (given Z full rank with finite second moments), so $v = \tilde{e}$ and $u = \delta\tilde{e} + w$ with $\tilde{e} \perp\!\!\!\perp w$—precisely the representation Proposition 5 rules out when at most one of u, e is Gaussian, and permits when both are. □

Lemma 6 is what licenses (7)–(8): the non-Gaussianity the construction needs is of the Z-residuals u and e, not of X_j and e_i themselves, and conditioning on Z leaves the bivariate argument intact only because Z enters both factorisations linearly and $e \perp\!\!\!\perp (X_j, Z)$ rather than merely $e \perp\!\!\!\perp X_j$.

DirectLiNGAM (Shimizu et al., 2011) and the pairwise measures of Hyvärinen and Smith (2013) exploit the first half by *comparing* the two directions' scores and taking the larger. Shimizu and Kano (2008) take a step towards inference by testing non-normality-based direction hypotheses in a structural equation modelling framework, and Komatsu, Shimizu, and Shimodaira (2010) assess LiNGAM's reliability by multi-scale bootstrap. What we add is a sequential e-process, which converts the comparison into a family-wise, time-uniform error statement over a whole graph.

5.2 The direction null and its e-process

Fix an ordered pair (i, j) whose adjacency has been certified, and a conditioning block Z (Section 5.5 explains how Z is chosen and why it is then frozen). The two competing factorisations of (X_i, X_j) given Z are

$$D^{j\to i}: \quad X_j = a^\top Z + u, \qquad X_i = \beta X_j + \gamma^\top Z + e, \tag{7}$$
$$D^{i\to j}: \quad X_i = b^\top Z + v, \qquad X_j = \delta X_i + \eta^\top Z + w, \tag{8}$$

with $u \perp\!\!\!\perp Z$, $e \perp\!\!\!\perp (X_j, Z)$ and correspondingly for the second. Both are conditional densities of the pair (X_i, X_j) given Z, so the marginal law of Z is a common factor and cancels from any ratio; this is why we may work with the conditional likelihoods and never model Z.

The certifying null for the arrowhead $i \to j$ is

$$H^{\text{dir}}_{j\to i}: \quad \text{the pair was generated by the } j \to i \text{ factorisation} = \{D_\theta^{j\to i} : \theta \in \Theta\}, \tag{9}$$

where $\theta = (a, \beta, \gamma, \text{noise parameters})$. Under Lemma 6 and non-Gaussianity of the Z-residuals u, e, $H^{\text{dir}}_{j\to i}$ is true whenever the arrowhead $i \to j$ is absent, so it is a certifying null in the sense of Definition 3.

We test it by *sequential universal inference* (Wasserman et al., 2020). Let $\hat{\theta}_{s-1}$ be any estimator of the alternative's parameters computed from $o_1, \dots, o_{s-1}$, and set

$$\log E_t \;=\; \sum_{s \le t} \log D^{i \to j}_{\hat{\theta}_{s-1}}(o_s) \;-\; \sup_{\theta \in \Theta} \sum_{s \le t} \log D^{j \to i}_{\theta}(o_s). \tag{10}$$

Proposition 7 *$(E_t)_{t \ge 0}$ defined by (10) is an e-process for $H^{\mathrm{dir}}_{j \to i}$, provided the supremum in the second term is attained (or exceeded) by the fitting procedure.*

Proof Let $P = D^{j \to i}_{\theta^\star} \in H^{\mathrm{dir}}_{j \to i}$ and define

$$M_t \;=\; \prod_{s \le t} \frac{D^{i \to j}_{\hat{\theta}_{s-1}}(o_s)}{D^{j \to i}_{\theta^\star}(o_s)}, \qquad M_0 = 1.$$

Since $\hat{\theta}_{s-1}$ is $\mathcal{F}_{s-1}$-measurable and $D^{i \to j}_{\hat{\theta}_{s-1}}(\cdot)$ is a probability density for each fixed value of $\hat{\theta}_{s-1}$,

$$\mathbb{E}_P\left[\frac{D^{i \to j}_{\hat{\theta}_{s-1}}(o_s)}{D^{j \to i}_{\theta^\star}(o_s)} \,\middle|\, \mathcal{F}_{s-1} \right] = \int D^{i \to j}_{\hat{\theta}_{s-1}}(o)\, do \;\le\; 1,$$

so (M_t) is a non-negative supermartingale (a martingale when the densities are mutually absolutely continuous) with $M_0 = 1$. The subtracted term in (10) is a supremum over Θ and $\theta^\star \in \Theta$, so it is at least $\sum_{s \le t} \log D^{j \to i}_{\theta^\star}(o_s)$, giving $E_t \le M_t$ pointwise. Optional stopping yields $\mathbb{E}_P[E_\tau] \le \mathbb{E}_P[M_\tau] \le 1$ for every stopping time τ. □

The argument is the standard universal-inference one, and it is worth being clear that its strength is also its cost. It requires no regularity conditions on Θ, no asymptotics, and no correctly specified alternative—the numerator may be fitted however one likes, since it only needs to integrate to one. In exchange, subtracting a maximised likelihood rather than an integrated one makes the resulting e-process conservative, typically by a factor related to the null model's dimension (Tse & Davison, 2022). Section 9.7 measures this: the certificate orients 88% of the edges DirectLiNGAM orients, which is the price of attaching an error statement to the answer.

5.3 Fitting the null, and the requirement that the supremum be real

We fit both factorisations with generalised Gaussian (exponential power) noise,

$$f_{\sigma,\kappa}(r) \;=\; \frac{\kappa}{2\sigma\,\Gamma(1/\kappa)} \exp\left\{ -\left| \frac{r}{\sigma} \right|^{\kappa} \right\}, \qquad \sigma > 0,\; \kappa > 0, \tag{11}$$

a family (Box & Tiao, 1973; Subbotin, 1923) whose shape parameter spans Laplace ($\kappa = 1$), Gaussian ($\kappa = 2$) and, as $\kappa \to \infty$, the uniform limit. Given κ, the coefficients are fitted by iteratively reweighted least squares on the $|r|^\kappa$ loss and σ is profiled out in closed form; κ is then chosen to maximise the profile likelihood.

Proposition 7 makes a demand on this last step that is easy to overlook, and we state it as a remark because we violated it.

Remark 4 (The supremum must be attained) The inequality $E_t \leq M_t$ in the proof holds because the subtracted term is at least the log-likelihood at $\theta^\star$. If the fitting procedure maximises over a strict subset of Θ that excludes $\theta^\star$, the subtracted term can fall *below* the log-likelihood at $\theta^\star$, E_t can exceed M_t, and the e-process property fails—silently, since nothing in the output signals it.

Our first implementation profiled κ over the fixed grid $\{0.7, 0.9, 1.2, 1.5, 2.0, 3.0, 5.0\}$, which omits $\kappa = 1$. On a correctly specified Laplace null, the fitted "supremum" fell below the likelihood at the true parameter in 32 of 200 replications, with a worst shortfall of 2.397 nats—an inflation of E_t by a factor of up to 11, and hence a certificate issued at a true level of up to 11α. The fix is to treat the grid as a bracketing device only and refine κ continuously by bounded scalar optimisation of the profile likelihood; the same check then gives 0 of 200 violations and zero shortfall. We report this because the defect is generic to universal inference with a discretised nuisance search, because it is invisible to a Type-I error simulation run at the same grid (the grid is wrong for the truth, not for itself), and because the correct check is cheap: simulate under the null at a known $\theta^\star$ and verify that the fitted supremum never falls below the likelihood there.

5.4 Abstention under Gaussianity

Proposition 8 *Suppose (X_i, X_j) given Z is jointly Gaussian, linear in Z. Then $H^{\text{dir}}_{j\to i}$ and $H^{\text{dir}}_{i\to j}$ both hold, and for every $\alpha \in (0,1)$ and every stopping time τ, the arrowhead $i \to j$ is certified at level α with probability at most α, and likewise for $j \to i$.*

Proof By the "both Gaussian" clause of Lemma 6, applied with u, e jointly Gaussian, (X_i, X_j) given Z admits *both* factorisations (7) and (8) for some parameter values, so $P \in H^{\text{dir}}_{j\to i} \cap H^{\text{dir}}_{i\to j}$. The proof of Proposition 7 did not use non-Gaussianity anywhere: it holds for every P in the null it names, whichever that is. Applying it to $H^{\text{dir}}_{j\to i}$, (E_t) built from (10) is an e-process for that null under joint Gaussianity, and Ville's inequality (1) gives $P(\exists\, t : E_t \geq 1/\alpha) \leq \alpha$, which is exactly the certification of $i \to j$. The symmetric argument, with the two factorisations' roles exchanged, bounds the certification of $j \to i$. □

The earlier version of this proposition claimed the deterministic statement $\log E_t \leq 0$ for every t, on the ground that the null and alternative models in (10) have identical maximised likelihoods under joint Gaussianity. That step does not go through: the numerator and denominator are each fitted over the generalised-Gaussian family of Eq. (11), whose member distributions coincide with each other only at the single point $\kappa = 2$, so their *sample* maximised log-likelihoods need not agree even when the data are exactly Gaussian—one family can fit an above-Gaussian-tailed realisation better than the other does, purely by chance. Simulating 300 replications of a conditionally Gaussian pair at $n = 400$ against the fitting procedure of Section 5.3, the reversed factorisation's maximised log-likelihood exceeded the forward one in 154 of 300 replications, by up to 3.8 nats—i.e. $\log E_t$ does occasionally exceed zero along the way. The revised proposition needs none of this: it does not claim the process never rises, only that crossing its threshold is no more likely than α, which is what Proposition 7 already gives for any null the pair satisfies, Gaussian or not.

The procedure therefore abstains exactly where the direction is not identified, at the same level α that governs every other certificate in this paper, rather than deterministically or because a Gaussianity pre-test told it to. This matters practically: a pre-test would itself have a Type-II error, and a method that orients whenever the pre-test fails to reject Gaussianity is once again asserting on non-rejection. Ding and Zhang (2025) give the underlying equivalence in a usable form—Gaussianity of the forward model's noise is equivalent to independence between regressor and residual in the *reverse* regression—and use it to avoid an explicit Gaussianity test in a different way. Section 9.7 shows the abstention empirically: under Gaussian noise the certificate orients 0 of 692 edges over 200 replications, while DirectLiNGAM orients all 692 and gets 356 of them wrong.

5.5 Freezing the conditioning block

The block Z in (7)–(8) is chosen when a pair's adjacency is certified: it is the set of that pair's already-certified neighbours at that moment. The direction process for the pair then begins accumulating, and Z is never changed.

For Eqs. (7)–(8) to be the correct linear non-Gaussian model for the pair, Z must satisfy two conditions, not one: it must contain every parent of both endpoints, *and* it must exclude every descendant of either endpoint. The first is the condition the implementation targets and Section 9.5 measures; the second is not targeted at all. Conditioning on a descendant of i or j—a collider on the path through that descendant, or a common effect of both—induces exactly the kind of dependence a linear non-Gaussian model does not have, for the same reason conditioning on a collider breaks faithfulness elsewhere in this literature. The certified-neighbour freeze does not rule this out: a variable certified adjacent to i or j before the pair itself is certified can be a child of one of them, not only a parent. Section 9.5's diagnostic therefore measures only half of what correctness of the frozen null requires, and its "complete"/"incomplete" label should be read as parent-completeness, not as a certificate that the direction null is correctly specified.

Freezing is not a convenience. A conditioning block that kept changing would change the hypothesis under test, and Ville's inequality bounds the crossing probability for a *fixed* null. There is a second, subtler issue: Z is chosen using the data, since it depends on which adjacencies have been certified. The resulting guarantee is therefore conditional on the σ-field $\mathcal{F}_T$ at the (stopping) time T at which the pair's adjacency is certified. Because the direction process uses only observations after T, and these are independent of $\mathcal{F}_T$ given the model, the supermartingale argument in Proposition 7 goes through conditionally on $\mathcal{F}_T$, and the unconditional bound follows by taking expectations. We make this precise in Appendix C. The consequence for the implementation is that the direction process must discard every observation before T, which costs power at pairs certified late.

Remark 5 (An implementation constraint that was a bug first) The observations scored by the numerator of (10) and those entering the denominator's maximised likelihood must be the *same* index set. A denominator ranging over additional observations subtracts their log-density with no matching numerator term, which adds to $\log E_t$ a quantity of order the number of extra observations. In our testing this certified *both* directions of every pair, including under Gaussian noise, where Proposition 8 guarantees the opposite. The symptom is diagnostic: whenever a direction procedure certifies both directions of a pair, the index sets are misaligned.

Remark 6 (Misspecification is the main theoretical risk) Proposition 7 needs the denominator's supremum to dominate the likelihood at the true null parameter. When the noise lies outside the fitted family (11), the supremum is taken over a model that does not contain the truth and no such domination is guaranteed. This is the certificate's principal theoretical gap, and unlike Remark 4 it cannot be closed by better optimisation.

Empirically the certificate is more robust than that suggests, and the original version of this check was too weak to say so: 0 of 12 runs at a single coefficient of 0.9—an easy design, since a coefficient that large makes the correct direction's evidence overwhelming regardless of the noise law—has a Wilson upper limit of 0.247, which cannot rule out a true error rate five times α. `scripts/check_misspecification_robustness.py` reruns the same three noise laws breaking each property of the family in turn—Student-t_3 (polynomial rather than exponential tails), shifted exponential (asymmetric) and a two-component mixture (bimodal)—at 150 replications per cell and a coefficient sweep of 0.9, 0.6 and 0.3, the last deliberately weakening the signal the earlier check left untested. The reversed direction was certified in 0 of 150 runs in every one of the nine (noise law, coefficient) cells, with a worst-case Wilson upper limit of 0.025 against $\alpha/2 = 0.025$—an order of magnitude tighter than the original check, and still zero at the weaker signal. But zero-count bounds remain bounds, not proofs, the study is bivariate, and Section 9.10 shows what happens on real data where the linear model itself is wrong: the certificate is confidently and systematically inverted. We regard bounding the misspecification gap analytically as the most important open problem the construction raises.

6 CERT-CD

6.1 The algorithm

CERT-CD maintains a three-valued status for every unordered pair and a two-valued status for every ordered pair. It streams data in batches. After each batch it updates the running Gram matrix, recomputes $A_t(i, j)$ for every undecided pair, and certifies adjacency when the running maximum of $A_t(i, j)$ crosses the pair's threshold. At that moment the pair's conditioning block is frozen and its two direction processes are opened. Certified arrowheads are added when a direction process crosses its own threshold.

Budgets are fixed before any data arrive. The total budget splits as $\alpha = \alpha_A + \alpha_D$; α_A is divided evenly over the $\binom{d}{2}$ adjacency nulls and α_D evenly over the $d(d-1)$ direction nulls, so the thresholds are

$$\lambda_A = \frac{\binom{d}{2}}{\alpha_A}, \qquad \lambda_D = \frac{d(d-1)}{\alpha_D}. \tag{12}$$

Algorithm 1 CERT-CD

Require: stream $o_1, o_2, \dots$; budget α; order k; warm-up m

$\alpha_A, \alpha_D \leftarrow$ split of α; λ_A, λ_D as in (12)
consume $o_1, \dots, o_m$ to fix standardisation constants and the prior scale g; **discard** them
status$(i,j) \leftarrow$ UNDECIDED for all pairs; arrow$(i,j) \leftarrow$ FALSE for all ordered pairs
$\bar{A}(i,j) \leftarrow 0$; $\bar{E}(i,j) \leftarrow 0$ ▷ running maxima
for each batch B **do**
    update the running Gram matrix with B
    for all pairs (i,j) with status(i,j) = UNDECIDED **do**
        $A \leftarrow \min_{S \in \mathcal{S}_k(i,j)} E^{(S)}(i,j)$ ▷ Schur complements of the Gram matrix
        $\bar{A}(i,j) \leftarrow \max\{\bar{A}(i,j), A\}$
        if $\bar{A}(i,j) \geq \lambda_A$ **then**
            status$(i,j) \leftarrow$ CERTIFIEDEDGE
            $Z_{ij} \leftarrow \{\, \ell : \text{status}(i,\ell) \text{ or status}(j,\ell) = \text{CERTIFIEDEDGE} \,\}$ ▷ frozen
            open direction processes for (i,j) and (j,i), starting from the next observation
        end if
    end for
    for all open ordered pairs (i,j) **do**
        update $\log E(i,j)$ by (10) on the observations since the pair was opened
        $\bar{E}(i,j) \leftarrow \max\{\bar{E}(i,j), E(i,j)\}$
        if $\bar{E}(i,j) \geq \lambda_D$ **then** arrow$(i,j) \leftarrow$ TRUE
        end if
    end for
    output the three-valued graph; the analyst may stop here
end for

Nothing about the split adapts to the data. Which certificates the algorithm happens to evaluate—it only opens direction processes for certified pairs—does not enter the bound, so the graph-dependent order in which pairs are examined is free.

The output graph contains only certified edges. An edge carries an arrowhead at an end where a direction certificate has fired and a circle otherwise, in the mark notation of Richardson and Spirtes (2002). **The absence of an edge means undecided, not certified absent.**

6.2 The whole-graph guarantee

Theorem 9 (Whole-graph anytime validity) *Suppose Assumptions 1 and 2 hold, the per-set processes $E^{(S)}$ are e-processes for their point nulls, and—for the direction certificates only—the pairs are generated by the linear non-Gaussian model of Section 5 with the supremum in (10) attained. Then*

$$\mathbb{P}\begin{pmatrix} \exists\, t \geq 0 : \textit{some edge certified by time } t \textit{ is absent} \\ \textit{from } \mathcal{G}^\star, \textit{ or some arrowhead certified by time } t \textit{ is wrong} \end{pmatrix} \leq \alpha. \tag{13}$$

In particular, for any *stopping time τ—including one chosen by inspecting the output—the graph reported at time τ satisfies $\mathbb{P}$(it contains a false certified edge or arrowhead) $\leq \alpha$.*

Proof Index the adjacency nulls by pairs and the direction nulls by ordered pairs; both families are determined by d and k and are fixed before any observation. Consider a single adjacency null for a pair (i, j) that is non-adjacent in $\mathcal{G}^\star$. By Lemma 2, $P \in H_{ij}^{\text{sep}}$; by Proposition 3, $(A_t(i, j))$ is an e-process for that null; so by (1) the probability that $A_t(i, j)$ ever reaches $\lambda_A = \binom{d}{2}/\alpha_A$ is at most $\alpha_A/\binom{d}{2}$. Since the algorithm certifies the edge exactly on that event, the probability that this pair is ever falsely certified is at most $\alpha_A/\binom{d}{2}$.

Similarly, consider an ordered pair (i, j) for which the arrowhead $i \to j$ is absent from $\mathcal{G}^\star$. Then $P \in H_{j\to i}^{\text{dir}}$, which by Proposition 7 (applied conditionally on $\mathcal{F}_T$, see Appendix C) has $(E_t(i, j))$ as an e-process, and the probability that it ever reaches λ_D is at most $\alpha_D/(d(d-1))$.

The event in (13) is the union, over the at most $\binom{d}{2}$ non-adjacent pairs and at most $d(d-1)$ absent arrowheads, of these events. A union bound gives $\binom{d}{2}\cdot\alpha_A/\binom{d}{2}+d(d-1)\cdot\alpha_D/(d(d-1)) = \alpha_A + \alpha_D = \alpha$.

The final statement follows because the event in (13) contains, for every stopping time τ, the event that the graph at time τ contains a false certified assertion; the bound is time-uniform, so no separate argument per stopping time is needed. □

We stress the form of the quantifier. The theorem does not say "for each fixed τ, the error at τ is at most α", which would be a family of pointwise statements requiring a further union bound to combine. It says that the probability of the trajectory *ever* containing a false certified assertion is at most α, and every stopping-time statement is a consequence. This is the difference between a fixed-sample guarantee applied repeatedly and a time-uniform one, and Section 9.2 measures how large that difference is in practice.

Remark 7 (What the theorem does not cover) Theorem 9 bounds the probability of a false *assertion*. It says nothing about the pairs left undecided, and it does not bound the probability that a true edge is missed—there is no such bound, and there cannot be one without a faithfulness-type assumption. It also does not cover the analyst's use of the output for anything else: an intervention chosen on the basis of a certified arrowhead inherits the α, but a downstream effect estimate computed from the certified graph does not automatically inherit anything, and would need its own analysis; the estimand there is the intervention-effect problem of Maathuis, Kalisch, and Bühlmann (2009), and combining a certified graph with a valid effect interval is a genuinely open question.

6.3 Complexity

Per batch, the Gram-matrix update is $O(d^2)$ per observation. Recomputing $A_t(i, j)$ for a pair costs $O(|\mathcal{S}_k| \cdot k^3)$, dominated by the Schur complements, and $|\mathcal{S}_k| = \sum_{r\le k}\binom{d-2}{r} = O(d^k)$. Summed over pairs, an update of the whole adjacency layer is $O(d^{k+2}k^3)$, with no dependence on n: all the data enter through the sufficient statistic. Only undecided pairs are updated, so the cost falls as the run proceeds.

The direction layer is the expensive part. Each open ordered pair requires a generalised-Gaussian fit at every batch, which is an IRLS loop plus a scalar optimisation over κ, and it does not reduce to a sufficient statistic because the fit is nonlinear in the residuals. The cost is $O(d^2 \cdot n_{\text{batch}} \cdot |Z| \cdot \text{iterations})$ per batch, and it is the reason our experiments stop at $d = 11$. This is an implementation limit, not a limit of the construction: the numerator's parameters may be refitted at any frequency (refitting less often only costs power, since the numerator needs to be $\mathcal{F}_{s-1}$-measurable and nothing more), and the denominator's maximised likelihood admits warm starts across batches. We have not pursued either optimisation and regard scaling as unproven; see Section 10.

6.4 On the multiplicity layer

Theorem 9 uses a union bound over a fixed family, and a union bound is the least powerful admissible way to combine evidence. It is reasonable to ask why we do not use a sharper procedure. E-value-based family-wise error control has advanced considerably: Hartog and Lei (2026) show that e-closed testing with weighted arithmetic-mean local tests strictly dominates e-Bonferroni, and their e-Holm procedure achieves this at $O(m \log m)$ cost. Crucially, the arithmetic-mean local test is valid under *arbitrary* dependence between the base e-values, so the fact that all our per-hypothesis processes are functions of the same running Gram matrix is *not* an obstacle. An earlier draft of this paper claimed it was; that claim was wrong, and we correct it here.

The actual obstruction is different and, we think, more interesting. Our certificates fire on the *running maximum* $\bar{A}_t = \sup_{s \leq t} A_s$ of an e-process, because that is what a monitored run rejects on. A running maximum is not an e-value. Ville's inequality bounds $\mathbb{P}(\bar{A}_\infty \geq 1/\alpha)$ by α, which makes $\bar{A}_t$ what Hartog and Lei (2026) call a *pseudo* e-value, but $\mathbb{E}[\bar{A}_\tau]$ generally exceeds one, and closed testing takes e-values as inputs, not pseudo e-values. Feeding running maxima into an e-closed procedure is not licensed by the theory. Hartog and Lei (2026) address this case directly and recover a bound of $\alpha + O(\alpha^2 \log \alpha^{-1})$ for pseudo e-values, but only under *independence* across hypotheses—and that is where our shared sufficient statistic does bite.

So the situation is: dependence is fine for e-closed testing with e-values; running maxima are fine under independence; we have running maxima under dependence, and no result covers the combination. The union bound over Ville events is what remains available, and it is exact for what it does. Closing this gap—a family-wise procedure for dependent pseudo e-values that improves on Bonferroni—would immediately improve the power of every method in this paper, and we regard it as the most valuable piece of the surrounding theory that is missing.

Two weaker improvements are available now and we note them. First, the budget split need not be uniform: any fixed weights $w_h > 0$ with $\sum_h w_h = 1$ give thresholds $1/(\alpha w_h)$ and the same bound, so prior knowledge about which pairs are likely adjacent can be spent as unequal weights, decided before the data arrive. Second, if the analyst wants false-discovery-rate rather than family-wise control, the e-BH procedure of Wang and Ramdas (2022) applies to e-values under arbitrary dependence, as do the martingale-based merging rules of Vovk and Wang (2024), and would replace the union bound directly—but only at a fixed stopping time, since it again takes e-values rather than running maxima as inputs.

6.5 An optional fourth verdict: certified negligible

Section 3 argued that there is no certifying null for non-adjacency. There is, however, one for a weaker proposition, and analysts who need it can have it at the cost of a stated margin. Fix $\delta > 0$ and consider the interval null

$$H^{\mathrm{big}}_{ij,S} \;:\quad |\rho_{ij\cdot S}| \geq \delta, \tag{14}$$

where $\rho_{ij\cdot S}$ is the partial correlation. Rejecting it for *every* S in a chosen set certifies that the pair's association is smaller than δ given each of those conditioning sets—a statement about effect size, not about the graph. The confidence sequence (6) supplies the test directly: reject when the interval excludes $[-\delta, \delta]$. The resulting verdict, *certified negligible at margin* δ, is honest in a way that a deleted edge is not, because δ is reported alongside it and the reader can judge whether an association below δ matters for their purpose. It is also strictly less than the claim a delete-on-non-rejection method makes, which is presumably why that claim is popular. We implement this verdict but do not use it in the experiments below, since the comparisons there are with methods that assert absence outright.

7 Latent confounders

The adjacency certificate does not require causal sufficiency; nothing in Lemma 2 assumes that all common causes are observed. What changes under latent confounding is the interpretation of a certified edge and the availability of the direction certificate.

If $\mathcal{G}^\star$ is a DAG over observed and latent variables and we observe only V, the relevant object is the maximal ancestral graph over V (Richardson & Spirtes, 2002), and the relevant separation notion is m-separation. Assumption 2 becomes the requirement that every non-adjacent pair in the MAG has an m-separating set of size at most k among the observed variables—which, unlike the DAG case, is not guaranteed by bounding an in-degree, and is why FCI searches Possible-D-SEP sets rather than neighbourhoods. With that caveat, Lemma 2 and Proposition 3 go through verbatim with d-separation replaced by m-separation, and a certified edge means "i and j are adjacent in the MAG", that is, neither is separable from the other by any observed set—which is compatible with a direct effect, a latent common cause, or both.

We therefore implement CERT-FCI: run the adjacency certificates to obtain a certified skeleton, apply the FCI orientation rules R1–R4 (originally Spirtes et al., 1995, in the formulation of J. Zhang, 2008) to the certified edges only, and report the resulting partial ancestral graph. Orientations produced by the FCI rules inherit the adjacency certificates' guarantee only in the weak sense that they are sound given a correct skeleton and correct sepsets; we do not claim a family-wise bound for them, and we flag this as the principal gap in the latent-variable extension. The direction certificate of Section 5 is unavailable here, because Proposition 5 assumes no unobserved common cause of the pair. Section 9.8 reports what CERT-FCI recovers, and Section 9.12 reports what latent confounding does to the direction certificate when it is applied anyway.

8 Related work

Most of the primitives used in this paper are standard, and the value of the contribution depends on saying precisely which. We therefore separate the accounting into what is not new, what we believe is new, and—because we think this is the more useful disclosure—three claims of novelty we made in earlier drafts and withdrew after reading the prior art.

8.1 What is not new

The minimum of e-values.

That $\min_k E^{(k)}$ is an e-value for a union null is standard (Vovk & Wang, 2021), and Proposition 3 is that fact applied to a particular union.

Edge-level aggregation over conditioning sets.

PC-p (Strobl et al., 2019) upper-bounds an edge's p-value by the *maximum* p-value over conditioning sets—the exact p-value analogue of (3)—and controls the false discovery rate of the resulting edge decisions with Benjamini–Yekutieli (Benjamini & Yekutieli, 2001). Treating adjacency rather than conditional independence as the object being tested is also the premise of DAT (Amin & Wilson, 2024), which replaces the exponential family of tests with a relaxed optimisation problem solved by neural networks.

One-sided, faithfulness-free false-edge control.

This is exactly Theorem 2 of ϵ-CUT (Shaska & Mitra, 2025), reached by the same reasoning we use: a guarantee against false edges needs a correct null, not faithfulness. We had believed this was ours; it is not.

Abstention and three-valued output.

FCI leaves unidentifiable edges undirected; the conservative PC of Ramsey et al. (2006) marks ambiguous triples rather than resolving them; Uehara (2026) attaches per-edge `resolved`/`impossible` codes with tiers that abstain when a precondition test rejects; and Prakash, Xia, and Erosheva (2024) give a bivariate direction test with an explicit inconclusive outcome.

Sequential e-values inside a constraint-based algorithm.

Csillag et al. (2025) already do this, calibrating batched Fisher-z p-values into per-test e-values within PC.

Residual-based direction criteria.

DirectLiNGAM (Shimizu et al., 2011) and the pairwise likelihood ratios of Hyvärinen and Smith (2013) use exactly the comparison in (10)'s two branches, and Ruiz, Madrid Padilla, and Zhou (2022) learn a topological ordering from likelihood-ratio scores on residuals. The entropy-based approximations underlying the pairwise measures go back to Hyvärinen (1998).

Orientation error control per se.

Strobl et al. (2019) control the FDR of orientations, using the rule set of Meek (1995a); Shimizu and Kano (2008) test direction of causation using non-normality; Komatsu et al. (2010) attach bootstrap reliability to LiNGAM outputs; Strieder and Drton (2023) give confidence statements under structure uncertainty for equal-variance models. None of these is a time-uniform family-wise statement, but the general project of attaching error control to orientation decisions is well established, and it would be wrong to present it as new.

8.2 Three withdrawn claims

We record these because the accounting above is only credible if the failures are visible too.

1. *"Orientation error is uncontrolled in the literature."* False. PC-p (Strobl et al., 2019) controls the FDR of unshielded-collider and Meek-rule orientations, and Shimizu and Kano (2008)—published in this journal's subject area two decades ago—tests direction of causation directly.
2. *"Faithfulness-free finite-sample false-edge control is new."* False. It is Theorem 2 of Shaska and Mitra (2025), at each fixed n.
3. *"Sequential anytime-valid e-values have not been used inside constraint-based discovery."* False; see Csillag et al. (2025).

8.3 What we believe is new

Three claims survive that accounting. We state each in the narrowest form we can defend.

(i) Orientation error control that is simultaneously finite-sample, time-uniform, family-wise over a graph, and not confined to the Markov equivalence class.

Each of these four properties has precedent separately. Strobl et al. (2019) is finite-sample and family-wise (in the FDR sense) but confined to the equivalence class, so an edge in no CPDAG is unorientable by it in principle. Shimizu and Kano (2008) escapes the class and is a genuine test, but is bivariate and fixed-sample. Ruiz et al. (2022) is closest in statistic—likelihood-ratio scores on residuals—but proves only population-level identifiability, requires all noise terms to come from a single scale-location family, and its "sequential" refers to greedy sorting rather than to sequential inference. Strieder and Drton (2023) is finite-sample and escapes the class but assumes equal error variances and targets causal effects rather than arrowheads. Uehara (2026) certifies *provenance*—which theorem licensed a direction—rather than error, and its Theorem 1 bounds expert query counts under an ideal-oracle assumption. We are not aware of a predecessor combining all four properties.

Table 1 Positioning against the closest prior work. "Beyond MEC" means the method can orient an edge that the CPDAG leaves undirected. "Time-uniform" means the error statement holds simultaneously at all sample sizes. "Faithfulness-free" refers to the *validity* of the method's assertions, not to its power. †CERT-CD's orientation FWER and faithfulness-freedom hold conditional on the linear non-Gaussian model describing the pair given its frozen block, a premise Section 9.5 measures failing on unfaithful instances; only the adjacency column is unconditional.

	error control on		beyond	time-	faithf.-
method	adjacency	orientation	MEC	uniform	free
PC (Spirtes et al., 2000)	–	–	no	no	no
PC-p (Strobl et al., 2019)	FDR	FDR	no	no	no
ϵ-CUT (Shaska & Mitra, 2025)	one-sided	–	no	no	yes
DAT (Amin & Wilson, 2024)	–	–	no	no	no
PPE-PC (Csillag et al., 2025)	per-test	–	no	yes	no
DirectLiNGAM (Shimizu et al., 2011)	–	–	yes	no	n/a
Shimizu and Kano (2008)	–	test	yes	no	n/a
Ruiz et al. (2022)	–	–	yes	no	n/a
Strieder and Drton (2023)	–	conf. set	yes	no	n/a
Uehara (2026)	provenance	provenance	no	no	no
CERT-CD (this paper)	FWER	FWER†	yes	yes	yes†

(ii) A whole-graph guarantee proved rather than assumed.

Per-test anytime-validity in discovery is not new (Csillag et al., 2025), but graph-level validity there is obtained by assumption: their Assumption 2.5 states that the downstream algorithm is valid whenever its inputs are, and they note the multiple-comparison concern without resolving it. Theorem 9 proves the statement over the fixed adjacency family unconditionally, and extends it to orientations only conditional on the linear non-Gaussian model describing each pair given its frozen block (Section 9.5 measures that premise failing on unfaithful instances). The proof is not deep—it is a union bound over Ville events—but the unconditional half of the statement is the one a practitioner needs for adjacency, and it was previously an assumption even there.

Table 1 sets the three claims against the closest prior work, one column per property.

(iii) Time-uniformity on the adjacency side.

This claim is a delta against a single unrefereed source: ϵ-CUT (Shaska & Mitra, 2025) is an arXiv preprint, not a peer-reviewed publication, at the time of writing, and a special issue's review lag is long enough that its status could change before this paper appears. ϵ-CUT holds the same one-sided, faithfulness-free ground *at each fixed* n. Bounding the probability that a false edge is *ever* certified over a monitored run is a different statement, and it is the one a monitored run needs; Section 9.2 measures the size of the gap. Secondarily, minimising over the complete family $\mathcal{S}_k$ performs all the tests that PC-p's maximum-p bound requires, so PC-p's zero-Type-II-error assumption is not needed here.

8.4 Other neighbouring literatures

Score-based and permutation-based discovery.

GES (Chickering, 2002) and greedy permutation methods (Solus, Wang, & Uhler, 2021) search over equivalence classes or orderings with consistency guarantees; they do not attach finite-sample error statements to individual features, and the certificate-only rule has no direct analogue there, since a score comparison is not a test.

Functional causal models beyond LiNGAM.

Additive noise models (Hoyer, Janzing, Mooij, Peters, & Schölkopf, 2009), post-nonlinear models (K. Zhang & Hyvärinen, 2009), causal additive models (Bühlmann, Peters, & Ernest, 2014) and equal-variance Gaussian models (Peters & Bühlmann, 2014) all break the equivalence class by restricting the functional form. Any of them could in principle replace the generalised-Gaussian likelihoods in (10), provided the resulting null model admits a computable supremum; this is the most direct route to relaxing Remark 6.

Invariance-based methods.

Invariant causal prediction (Peters, Bühlmann, & Meinshausen, 2016) identifies a set of causal parents by intersecting the environments in which a candidate set is invariant, and controls the probability that the returned set is not a subset of the true parents. The logic is close to ours in spirit—it asserts by rejecting non-invariance, and returns the empty set rather than guessing—and it is the one established method whose guarantee has the same one-sided shape. Section 9.9 reports an anytime-valid variant of it built on the same primitives.

Terminology.

"Anytime FCI" (Spirtes, 2001) denotes a *computational* property: the outer loop over conditioning-set sizes may be interrupted and the algorithm still returns a sound graph. "Anytime-valid" in this paper is an *inferential* property: time-uniform error control at data-dependent stopping times. The two notions are unrelated, and the collision of names is unfortunate.

9 Experiments

9.1 Protocol

Every experiment streams data in batches and inspects the estimate after every batch, scoring the *worst case over the whole trajectory* rather than the state at a single sample size. This is deliberate: a time-uniform guarantee is a claim about trajectories, and evaluating it at a single n would measure something else. Where a comparator has only a fixed-sample guarantee we report both its trajectory-worst behaviour (which is what an analyst monitoring it would experience) and its behaviour at the final sample size (which is what its guarantee covers), because reporting only the first would be unfair and reporting only the second would miss the point.

Table 2 Probability of ever rejecting a true null over a monitored run (2000 replications, $n \leq 1000$), with Wilson 95% intervals. "Fisher-z, monitored" inspects after every observation; "Fisher-z, once" evaluates only at $n = 1000$.

nominal α	Fisher-z, monitored	Fisher-z, once	e-process (monitored)
0.01	0.100 [0.088, 0.114]	0.010 [0.006, 0.015]	0.007 [0.004, 0.012]
0.05	0.346 [0.325, 0.367]	0.056 [0.046, 0.066]	0.024 [0.018, 0.031]
0.10	0.565 [0.543, 0.586]	0.103 [0.090, 0.117]	0.049 [0.040, 0.059]
0.20	0.816 [0.798, 0.832]	0.193 [0.176, 0.211]	0.091 [0.079, 0.104]

Random DAGs are drawn with independent edge probabilities in a random topological order, with coefficients uniform on $\pm[0.5, 1.2]$. Noise is Laplace unless stated. Binomial proportions carry Wilson 95% intervals. Code, seeds and figures are available in the repository (Section 11); every number below is reproduced by a script in `experiments/`.

Two comparators recur. **PC** is the standard algorithm with a Fisher-z test, re-run from scratch at each sample size on the same stream. **SPRINT-CD** is an anytime-valid PC variant we implemented ourselves for this comparison: it deletes an edge when a confidence sequence for the partial correlation falls entirely inside a region of practical equivalence. It is unpublished, and we make no claim that it is the strongest possible delete-on-non-rejection design with sequential machinery—only that it lets the comparison isolate the deletion rule from the fixed-sample/sequential distinction, since it shares CERT-CD's streaming primitives and differs only in deciding absence by failing to reject rather than by never asserting without rejecting. A published sequential deletion rule, were one available to compare against, could behave differently. Both PC and SPRINT-CD decide absence by failing to find evidence of presence; that is the property under test.

9.2 What monitoring costs a fixed-sample test

Data are generated under a true conditional independence and the analyst inspects the evidence as observations accumulate. Over 2000 replications monitored to $n = 1000$, Table 2 reports the probability that each procedure ever rejects.

The middle column confirms that the Fisher-z test is correctly calibrated for the guarantee it makes; the first column shows that the guarantee does not survive looking. At $\alpha = 0.05$ the realised error rate under monitoring is 0.346, seven times nominal. The e-process stays below nominal throughout—conservatively so, which is the other half of the story.

The price in power.

Time-uniformity is not free, and we report the cost directly. Against a fixed-sample Fisher-z test evaluated once at $n = 1000$, over 500 replications per setting, the e-process detects a coefficient of $\beta = 0.04$ in 5% of runs against 21%; $\beta = 0.06$ in 16% against 47%; $\beta = 0.08$ in 32% against 72%; $\beta = 0.10$ in 58% against 91%; and $\beta = 0.15$ in 96% against 100%. Median stopping times where the e-process does fire range from 360 to 520 observations. **When the sample size can honestly be fixed**

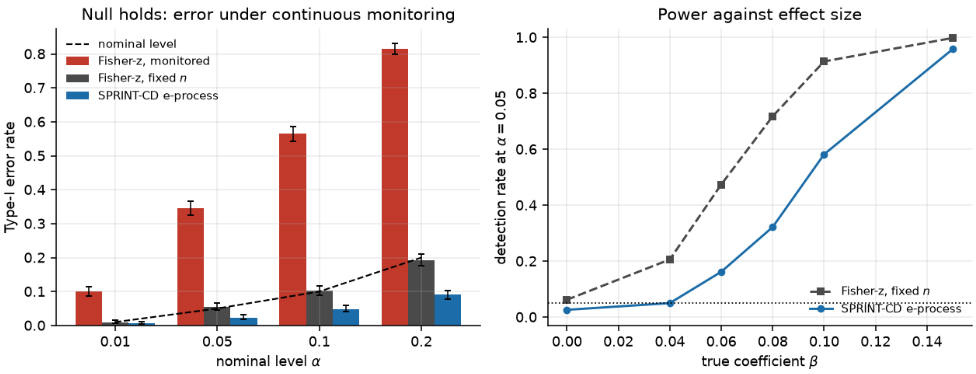

Fig. 1 Left: probability of ever rejecting a true null under continuous monitoring, against nominal level. Right: power against a fixed-sample test at the same n. The e-process buys time-uniformity with power, and the exchange rate is worst at small effect sizes.

in advance, a fixed-sample test is the better instrument, and by a wide margin at small effect sizes. The e-process earns its keep only when the analyst intends to look, which in a discovery setting is usually. Figure 1 shows both panels.

9.3 The primitive under six noise laws

Proposition 4 assumes Gaussian noise. Since the adjacency guarantee inherits whatever the primitive assumes (Remark 1), we measured the primitive's calibration under six noise laws—Gaussian, Laplace, uniform, Student-t_5, shifted exponential and a two-component mixture—over 2000 replications each, monitored to $n = 800$. Realised crossing rates never exceeded nominal: the largest ratio of realised to nominal anywhere in the table is 0.95, at shifted exponential noise and $\alpha = 0.01$, where 19 of 2000 runs crossed against an allowance of 20; the median ratio is 0.49. The Gaussian and Laplace columns agree to within Monte Carlo error. Full numbers are in Appendix D, Table D1, and Figure 2 plots them.

This does not prove validity outside the Gaussian model—it is six laws, not a theorem—but it does bound the practical risk from non-Gaussian noise specifically. It says nothing about a different way the primitive can be wrong: Eq. (4) assumes the conditional mean is linear, and none of the six noise laws breaks that assumption, only the error distribution around it. Section 9.10 certifies edges on data we ourselves describe as demonstrably nonlinear, and we have not measured what nonlinearity of this kind does to the adjacency primitive's calibration; the six-noise-law result should not be read as ruling that risk out.

9.4 Measuring our own premise

Assumption 2 replaces faithfulness, and it would be self-serving to compare a method that assumes one to a method that assumes the other without saying how often each holds. Both are properties of the generating model, computable before any data are

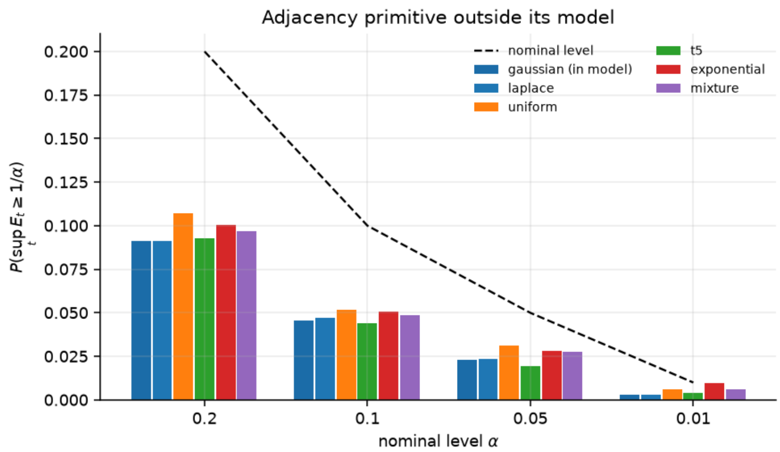

Fig. 2 Realised versus nominal crossing rates for the per-set primitive under six noise laws, 2000 replications each. The dashed line is $y = x$.

Table 3 How often each method's premise holds, over 2000 random DAGs per setting. Assumption 2 is checked directly by d-separation, not via the sufficient in-degree condition. "Adjacency-faithful" restricts the 0.15 margin to pairs adjacent in the true skeleton, the weaker condition an earlier version of this table reported under the "0.15-strong faithfulness" label; the column beside it is the condition Section 2.3 actually defines, quantified over every pair. The last row marginalises one variable out of a 7-node graph and additionally requires the latent to confound at least two observed edges, matching Section 9.8, so the observed model has a genuine confounder.

setting	Assumption 2 at $k = 2$	adj.-faithful	0.15-strong faithful
$d = 5$, edge prob. 0.35	0.988	0.758	0.581
$d = 6$, edge prob. 0.30	0.972	0.663	0.404
$d = 7$, edge prob. 0.35	0.815	0.325	0.102
$d = 7$, one latent, confounding	—	—	0.054

simulated, so the comparison is cheap. Over 2000 random DAGs per setting, Table 3 reports how often each premise holds at $k = 2$ and $\delta = 0.15$.

One definitional point matters here and an earlier version of this paper got it wrong. δ-strong faithfulness constrains *every* partial correlation not forced to zero by d-separation, over all pairs. A margin computed only over pairs that are *adjacent* in the true skeleton is the weaker adjacency-faithfulness condition, and reporting it as δ-strong faithfulness overstates how often the competing premise holds—in the direction that flatters the comparators. The column below is the full condition, truncated at $|S| \leq k$ because that is the family the experiments test; the adjacency-only margin is reported beside it so the gap is visible.

Two things follow. First, our premise is substantially more often satisfied than the one it replaces, and by a widening margin as the graph grows: at $d = 7$ with a confounding latent variable, 0.15-strong faithfulness holds in about one instance in twenty. Second—and this is the reason to report the table rather than assert the first

Table 4 CERT-CD's own two error types: probability of ever making an unsound assertion at any point in a monitored run, at $\alpha = 0.1$, $d = 5$, Laplace noise, $n \leq 3000$, with Wilson 95% intervals. Parenthesised counts are numbers of *instances* in each stratum, not sample sizes. The budget is split evenly, so each column is bounded by $\alpha_A = \alpha_D = 0.05$, not by 0.1. Both quantities are trajectory-worst.

	CERT-CD	
stratum	false edge	wrong arrow
δ-strong faithfulness holds (500 instances)	0.000 [.000,.008]	0.004 [.001,.015]
δ-strong faithfulness fails (250 instances)	0.016 [.006,.040]	**0.080** [.052,.120]

Table 5 The comparators' own error type on the same instances as Table 4: probability of ever losing a true edge at any point in the monitored run (trajectory-worst), same setting and Wilson intervals. SPRINT-CD and PC both decide absence by failing to reject, so a lost edge is their analogue of an unsound assertion; CERT-CD never deletes, so it has no entry in this table and Table 4 has none in this one. The two tables measure different failure modes on a shared set of instances—they are not meant to be read column-against-column, and no ratio taken across them is a validity comparison; only the within-table pattern by stratum supports one.

stratum	SPRINT-CD lost edge	PC lost edge
δ-strong faithfulness holds (500 instances)	0.000 [.000,.008]	0.008 [.003,.020]
δ-strong faithfulness fails (250 instances)	0.560 [.498,.620]	0.812 [.759,.856]

point—Assumption 2 is *not* always satisfied either, and its violation is just as silent as unfaithfulness: if a non-adjacent pair has no separating set within $\mathcal{S}_k$, the separability null is false for that pair, (A_t) is not an e-process for it, and a false edge can be certified with no bound at all. Section 9.12 measures what happens then.

9.5 Certificate-only against delete-on-non-rejection

This is the central comparison. Instances are stratified by whether 0.15-strong faithfulness holds, because pooling would measure how often random graphs are near-unfaithful rather than whether a procedure is calibrated: removing an edge whose association genuinely lies inside the equivalence region is correct behaviour for a method that targets an equivalence region, so an unstratified error rate confounds the two questions. Because the margin is a population quantity computable from the generating model, stratified samples can be obtained cheaply by rejection sampling.

Over 948 random DAGs, 0.15-strong faithfulness held in 73.6%; Table 4 and Table 5 report the two strata separately at 500 and 250 instances, CERT-CD's own error types in the first and the comparators' in the second, because a false edge or a wrong arrow is not the same failure as a lost edge and the two are not a single table's columns.

Where the premise holds, all four procedures behave. Where it fails—the regime that breaks delete-on-non-rejection—SPRINT-CD loses a true edge in 56% of runs and PC in 81% (Table 5), while CERT-CD's adjacency certificates stay comfortably inside their $\alpha_A = 0.05$ budget at 0.016 ($[0.006, 0.040]$, Table 4). Near-unfaithful instances cost the adjacency layer power rather than validity: the affected pairs stay undecided instead of being asserted wrongly. This is a within-stratum contrast for each method against its own budget or its own failing behaviour, not a claim that 0.016 is smaller than 0.560 in any sense that makes CERT-CD's adjacency layer superior to SPRINT-CD's skeleton on a shared scale—the two numbers answer different questions.

The orientation budget is exceeded on the failing stratum.

The bolded entry in Table 4 is a wrong arrowhead in 20 of 250 runs, 0.080 with interval $[0.052, 0.120]$, against an orientation budget of $\alpha_D = 0.05$. The interval excludes the budget, so this is not sampling noise: on premise-failing instances the direction certificates do not deliver the guarantee of Theorem 9. A lower-replication version of this experiment reported 0/20 here and we drew the opposite conclusion from it; the effect is only visible at a few hundred instances per stratum.

The mechanism is the one Remark 6 names, and it reaches the direction certificate through the adjacency layer rather than independently of it. The conditioning block Z_{ij} is frozen to the pair's *already-certified* neighbours (Section 5.5). On a near-unfaithful instance fewer neighbours have been certified by the time a pair is opened, so Z_{ij} more often omits a variable that is a parent of i or of j. The pair then has an uncontrolled common cause or an omitted direct parent, the linear non-Gaussian model of Section 5 is not the truth, and the domination step in the proof of Proposition 7 has nothing to stand on.

We tested that explanation rather than asserting it. Over 250 instances in each stratum we recorded, for every certified arrowhead, whether the frozen block Z_{ij} contained every parent of i and of j, and whether the arrowhead was wrong. Table 6 reports the result. Note that these are per-*arrowhead* rates, whereas Table 4 reports per-*run* rates.

As Section 5.5 states, a correctly specified frozen null needs Z_{ij} to satisfy two conditions, and this diagnostic checks only one: it labels a block "complete" whenever every parent of both endpoints is present, but it cannot see whether Z_{ij} also contains a descendant of either endpoint, which would misspecify the null independently of parent completeness. "Complete" below therefore means parent-complete, not correctly specified, and the low error rate in the complete cells is evidence about the parent-omission mechanism specifically, not a general certificate that the null is right whenever this diagnostic calls it complete.

Both halves of the explanation hold. Incompleteness is more common where the premise fails—49.0% of certified arrowheads against 31.5% where it holds (Fisher $p = 1.4 \times 10^{-12}$)—which is the certification-order effect. And incompleteness is where the

Table 6 Wrong-arrowhead rate by whether the frozen conditioning block was parent-complete (contained every parent of both endpoints; descendant membership is not checked and could misspecify the null in the parent-complete cells too), 250 instances per stratum, Wilson 95% intervals. Rates are per certified arrowhead.

stratum	Z_{ij}	arrowheads	wrong	rate
premise holds	complete	486	2	0.004 [.001,.015]
premise holds	incomplete	224	3	0.013 [.005,.039]
premise fails	complete	465	3	0.007 [.002,.019]
premise fails	incomplete	447	22	0.049 [.033,.073]

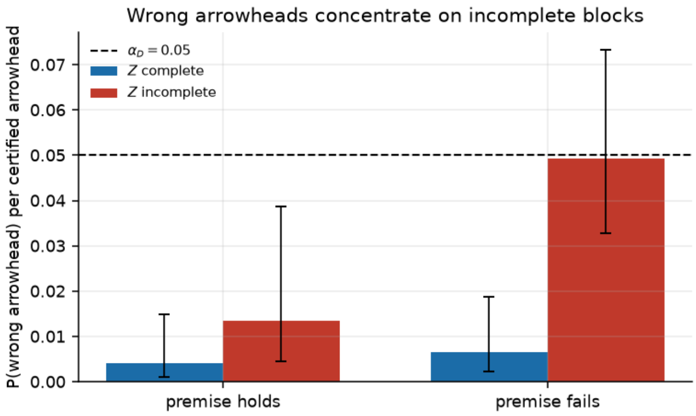

Fig. 3 Wrong-arrowhead rate by conditioning-block completeness, with Wilson 95% intervals. The dashed line marks α_D's numerical value for scale only: α_D bounds the whole-run, family-wise probability of any wrong arrowhead over all $d(d-1)$ direction nulls, not the per-arrowhead rate plotted here, so a bar falling below the line is not a budget statement. The exceedance in Table 4 lives entirely in the incomplete-block bar on the premise-failing stratum.

errors are: on the failing stratum the wrong-arrowhead rate is 0.049 with an incomplete block against 0.007 with a complete one, a factor of 7.6 (Fisher $p = 5.5 \times 10^{-5}$), and 22 of the 25 wrong arrowheads occur on pairs with an incomplete block. These are per-arrowhead rates, not the quantity α_D bounds: α_D is a whole-run, family-wise budget summed over all $d(d-1)$ direction nulls (Section 6.1), so the two are not on a common scale and the per-arrowhead rate being numerically smaller than α_D is not itself a statement that any budget is respected. What the stratification does support is the qualitative claim already made: the errors are not spread evenly but concentrated on the incomplete-block cells, which is where Theorem 9's direction premise is violated.

Figure 3 plots the four cells at the same numerical scale as α_D, without claiming the two are the same quantity.

The exceedance in Table 4 is therefore not a failure of Proposition 7 or of its implementation. It is the misspecification gap of Remark 6 being opened by a specific, identifiable mechanism, and it is reached through the algorithm's own scheduling rather than through anything intrinsic to the direction null.

Two consequences should be stated plainly. First, Theorem 9's direction clause is conditional on a premise—causal sufficiency for the pair *given the frozen block*—that the algorithm cannot verify and that is correlated with faithfulness through the certification order. It is therefore not the assumption-light guarantee the adjacency clause is, and the whole-graph bound holds unconditionally only for adjacency, as Section 1 now states.

Second, there is no cheap fix, and we record why rather than gesturing at one. The obvious alternative—freezing Z_{ij} to *all* other variables rather than to the certified neighbours, removing the dependence on certification order—trades one failure for a worse one: with no orientations available at freezing time, an all-variables block cannot exclude a collider or a descendant of either endpoint, and conditioning on either takes the pair outside *both* branches of Eqs. (7)–(8), so the certifying-null property fails even when every true parent is present and the global model holds everywhere. The certified-neighbour freeze used throughout this paper can miss a true parent; the all-variables freeze can include a collider; neither is a correct default, and we are not aware of a rule for Z_{ij} that avoids both failure modes with only the orientations available at the moment a pair is certified. Closing this is, alongside the multiplicity gap of Section 6.4, the open problem we consider most pressing.

Figure 4 shows both panels: the validity comparison by stratum, and what accumulates as n grows.

What is certified.

On premise-satisfying instances at $n = 3000$, 97% of true edges are certified and 93% are also oriented—so 96% of *certified* edges carry a direction—against a Markov-equivalence-class orientation ceiling of 33%, the fraction a CPDAG can orient at all on these graphs. That gap is the practical content of the direction certificate: an edge left undirected in the CPDAG is unorientable in principle by any constraint-based method, and roughly two thirds of these edges are of that kind.

The cost side of the same run is that 72% of all pairs remain undecided at $n = 3000$. Most of those pairs are genuinely non-adjacent—the graphs are sparse—and CERT-CD is structurally unable to say so. A reader comparing output sizes should keep this in view: PC returns a complete verdict on all 10 pairs, CERT-CD returns a verdict on about three.

9.6 Skeleton recovery and sample efficiency

The comparison above is at $d = 5$ with orientation enabled. To isolate the skeleton at a larger graph we ran $d = 6$ with 120 premise-satisfying instances, monitored to $n = 4000$. Here the quantity of interest for the comparators is a *lost* edge—an edge of the true CPDAG that the method deletes at some point in the run—since that is the error mode that delete-on-non-rejection has and CERT-CD structurally cannot have.

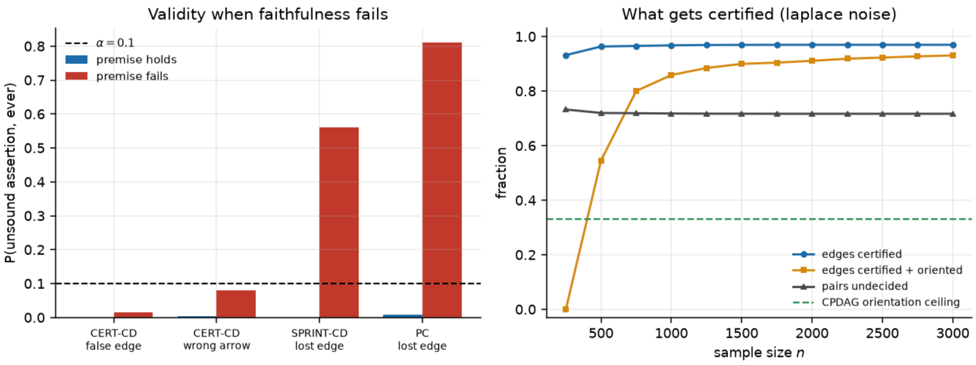

Fig. 4 Left: probability of an unsound assertion at any point in a monitored run, by stratum. Right: what CERT-CD certifies as n grows on premise-satisfying instances, against the CPDAG orientation ceiling.

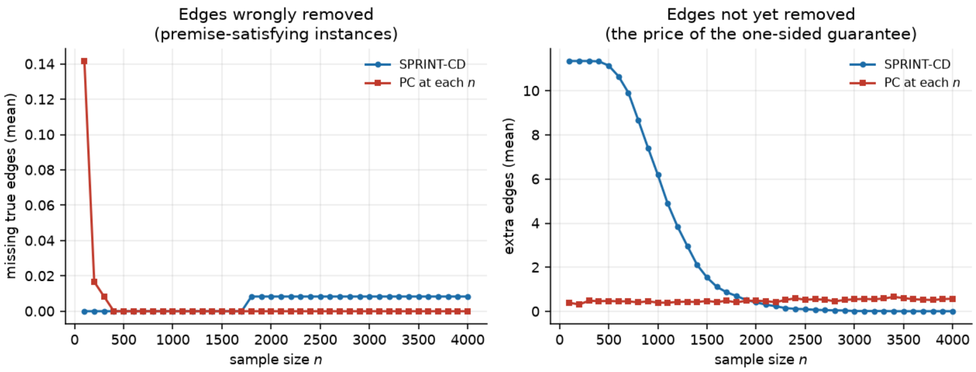

Fig. 5 Skeleton errors over monitored runs at $d = 6$. SPRINT-CD's extra edges fall from 12.2 to 0.00 as evidence accumulates, while its missing edges stay at zero on premise-satisfying instances.

Over the 285 DAGs generated for this stratification, 0.15-strong faithfulness (the condition of Section 2.3, not the adjacency-restricted margin of Table 3's earlier printing) holds in 42%. SPRINT-CD ever loses a true edge in 0 of 120 premise-satisfying runs (0.000, Wilson 95% interval $[0.000, 0.031]$) against a budget of 0.1, and in 14 of 40 premise-violating runs (0.350, $[0.221, 0.505]$). PC re-run at each sample size on the same premise-satisfying instances ever loses one in 5 of 120 runs (0.042, $[0.018, 0.094]$)—inside the 0.1 level here, though Section 9.2 shows the same monitoring effect can push it above budget at other settings. These are SPRINT-CD's and PC's numbers, not CERT-CD's: CERT-CD never deletes, so "lost edge" is not defined for it, and the corresponding quantity—a pair that is truly adjacent and never certified—is a power statement, reported in Section 9.5. Figure 5 shows the trajectories.

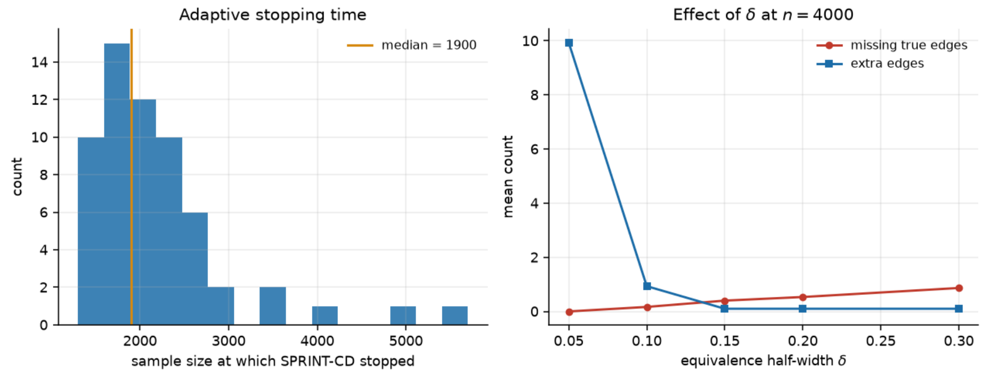

Fig. 6 Left: distribution of data-dependent stopping times and the resulting structural Hamming distance, against PC evaluated at fixed sample sizes. Right: sensitivity to the equivalence-region width.

Table 7 Orientation over 200 replications at $d = 5$, $n = 3000$, $\alpha = 0.1$. "Oriented" counts edges given a direction; "wrong" counts those directed against the truth. Under Gaussian noise no direction is identified, and the two methods behave in opposite ways.

	CERT-CD direction certificate		DirectLiNGAM	
noise	oriented	wrong	oriented	wrong
Laplace	609/689	3	689/689	0
Gaussian	0/692	0	692/692	356

Sample efficiency is reported separately (Figure 6). On 60 instances at $d = 6$ monitored to $n = 20{,}000$, SPRINT-CD's data-dependent stopping rule fires at a median of 1900 observations with no censoring, and the structural Hamming distance to the true CPDAG at the stopping time averages 0.83 (standard error 0.22), against PC's 1.20 at a fixed $n = 2000$ and 0.90 at $n = 8000$. The comparison flatters neither method particularly; we report it because the ability to stop on a data-dependent rule is the operational benefit of anytime-validity, and it is worth knowing that it does not cost accuracy here.

9.7 Orientation against DirectLiNGAM

The direction certificate's natural comparator is DirectLiNGAM (Shimizu et al., 2011), which shares the linear non-Gaussian model and the pairwise likelihood-ratio statistic but takes the larger score rather than testing. Over 200 replications at $d = 5$, $n = 3000$, Table 7 reports both under Laplace and Gaussian noise.

Under Laplace noise the certificate orients 88% of the edges DirectLiNGAM orients and gets 3 wrong out of 609, in 3 of 200 runs—inside the $\alpha = 0.1$ family-wise

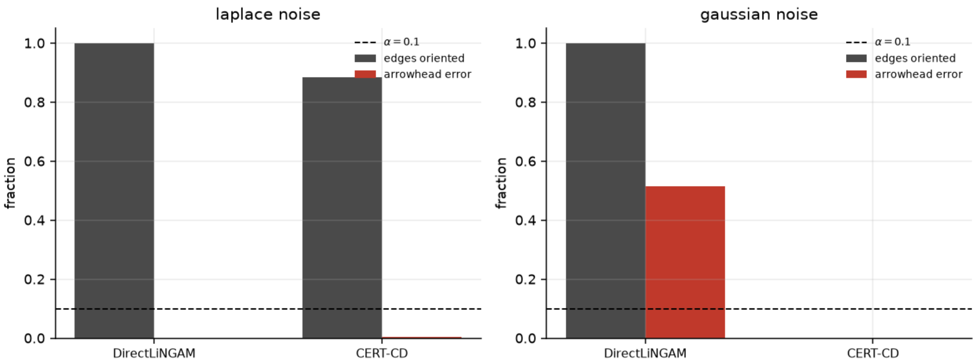

Fig. 7 Orientation under Laplace and Gaussian noise. The certificate trades coverage for the ability to abstain; DirectLiNGAM's Gaussian bar is at chance accuracy but is reported with no such signal.

budget, and the 12% shortfall in coverage is the conservatism of universal inference (Section 5.2) made concrete. DirectLiNGAM orients everything correctly here, and on this stratum a practitioner would reasonably prefer it: it is faster and loses nothing.

The Gaussian row is the reason not to. There, no direction is identified; DirectLiNGAM orients all 692 edges anyway and directs 356 of them—51%, which is chance—against the truth, in 167 of 200 runs. The certificate orients none, as Proposition 8 requires. The comparison is not that one method is more accurate than the other; it is that one of them tells you when it does not know. Figure 7 shows both noise regimes side by side.

9.8 Latent confounders

A note on what was run. An earlier version of this experiment reported numbers produced by the delete-on-non-rejection comparator under the CERT-FCI label; the metrics gave it away, since "lost adjacency" is not a defined error mode for a procedure that never deletes. CERT-FCI is now implemented as Section 7 describes it—an add-only certified skeleton with the FCI rules applied on top—and the two procedures are reported separately, each against the error mode it actually has.

Both CERT-FCI and its delete-on-non-rejection comparator SprintFCI were run on 7-variable graphs with one confounding variable marginalised out, $\alpha = 0.1$, monitored to $n = 4000$. CERT-FCI never certified an edge absent from the oracle skeleton, in either stratum (0/80 premise-satisfying, 0/40 premise-violating): its adjacency layer is unchanged from Section 4 and inherits that validity regardless of the faithfulness stratum. What varies with the stratum is power, not soundness: on the 80 premise-satisfying instances CERT-FCI certifies every oracle adjacency by $n = 4000$ and leaves 74% of pairs undecided at that size—the add-only layer is conservative here, and the undecided fraction is the cost of that.

SprintFCI, which does delete, is the one for which "lost adjacency" is a defined error mode. It never loses one on the 80 premise-satisfying instances (0/80) but loses one in 26 of 40 premise-violating instances (0.650). We report this stratum explicitly because

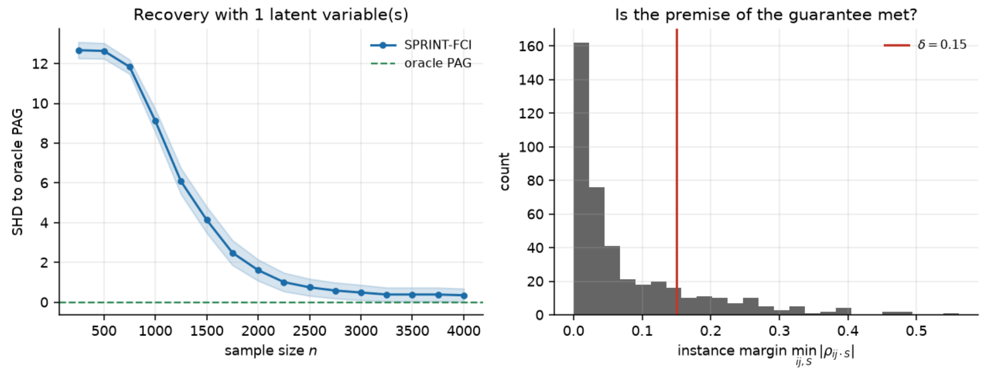

Fig. 8 CERT-FCI under one marginalised latent variable: structural Hamming distance to the oracle PAG against sample size, and the distribution of δ-strong faithfulness margins showing how few instances satisfy the competing premise.

pooling would flatter the comparator: at $d = 7$ with a confounding latent variable, 0.15-strong faithfulness now holds in only about one instance in twenty (Table 3), so the premise-satisfying stratum is a small minority of graphs drawn at random, not the typical case. Structural Hamming distance of SprintFCI's estimated PAG to the oracle falls from 12.3 to 0.01 on the premise-satisfying stratum, and the single instance with a bi-directed oracle edge has it recovered. Figure 8 shows the trajectory and the margin distribution.

9.9 An anytime-valid invariant predictor

Invariant causal prediction (Peters et al., 2016) shares the one-sided logic of Section 3: it accepts a candidate parent set only if invariance across environments is not rejected, and returns the intersection of the accepted sets, which is a subset of the true parents with high probability. Its guarantee is fixed-sample, so it degrades under monitoring in the same way Table 2 shows. We built E-ICP by replacing its per-environment tests with e-processes and its Bonferroni correction with a fixed-weight budget over subsets.

Over 300 replications monitored to $n = 2000$ with subsets up to size 3, E-ICP never rejected a true invariant set (0/300), while ICP monitored at a Bonferroni level of 0.0125 rejected in 24 runs and ICP evaluated once rejected in 2. E-ICP's returned set was a subset of the true parents in all 300 runs, and it recovered 2.0 of 2 true parents on average against ICP's 1.99. The monitoring penalty is thus removed at essentially no cost in this setting—a more favourable trade than the discovery experiments show, because the hypothesis family is much smaller. Figure 9 shows both quantities.

9.10 Real data: protein signalling

Every result so far comes from a simulation whose generating model is the one the certificates assume. We therefore ran CERT-CD on the single-cell protein-signalling data of Sachs et al. (2005), using the observational condition only—853 cells, 11

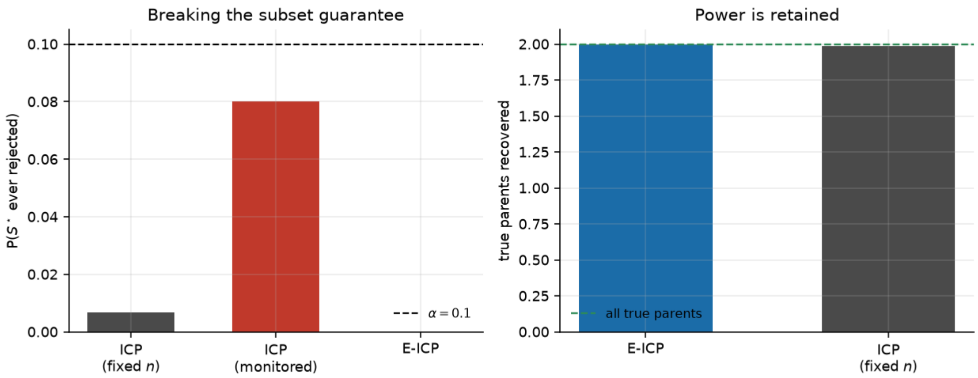

Fig. 9 E-ICP against ICP under continuous monitoring: rejection of a true invariant set, and recovery of the true parent set.

phosphoproteins, in the continuous form distributed with `bnlearn` (Scutari, 2010). Measurements are log-transformed, as in the original analysis, since the raw fluorescence intensities span three orders of magnitude and are strongly right-skewed. The interventional conditions that make the Sachs network identifiable are withheld, which makes this the hardest honest setting for a purely observational method. We compare against the published 17-arc consensus network, and we stress that this is a scientific consensus assembled from interventional experiments and prior literature, not ground truth; several of its arcs are contested.

At $\alpha = 0.1$ and $k = 2$, streaming in batches of 50 after a 100-observation warm-up that is discarded—so the e-processes see 753 of the 853 cells—CERT-CD certifies 6 edges out of 55 pairs and leaves 49 undecided. The two halves of the output behave very differently, and the contrast is the reason to report the experiment.

The adjacency half does what it claims.

All 6 certified edges lie in the consensus skeleton, and none outside it: Raf–Mek, PIP2–PIP3, Erk–Akt, Akt–PKA, PKC–P38 and PKC–Jnk. Six of 17 consensus edges is low recall, which is what one expects from the 753 observations the certificates actually consume, a family-wise budget over 55 pairs, and a certificate that must reject to assert. PC at the same level and order returns 7 edges, all also in the consensus skeleton, and then *calls the remaining* 48 *pairs absent*—an assertion about 48 pairs that this sample cannot support, and the one CERT-CD declines to make. The difference between the two outputs is not the 6 versus 7 edges; it is the 49 pairs one of them describes accurately.

The orientation half is confidently wrong.

All 4 certified arrowheads are reversed relative to the consensus: the certificate reports Mek $\to$ Raf, PIP2 $\to$ PIP3, Akt $\to$ Erk and P38 $\to$ PKC, against consensus arcs running the other way. Four out of four is not sampling variation, and we investigated it rather than reporting it as an anomaly.

The reversal is a property of these data under a linear non-Gaussian model, not of an idiosyncrasy of the sequential machinery. DirectLiNGAM produces a causal order that reverses *the same four arcs* (and agrees with the consensus on Akt–PKA and PKC–Jnk, which the certificate left unoriented). We do not read this agreement as independent corroboration: DirectLiNGAM uses exactly the pairwise comparison of Eq. (10)'s two branches (Section 8), so the two procedures share their statistic and their model, and agreeing on all four reversals is what two implementations of the same comparison should do when the shared model is wrong, not evidence that anything beyond the model has been ruled out. What the agreement does show is that the reversal is not an artefact of the sequential apparatus—the plain comparison reverses the same arcs with no e-process, no threshold and no monitoring involved. The pairs in question are embedded in a densely confounded, demonstrably nonlinear signalling network, where the bivariate linear non-Gaussian model of Section 5 is simply false. Under misspecification the supremum in (10) is taken over a model not containing the truth, and Proposition 7 gives nothing (Remark 6).

We draw two conclusions and resist a third, and we are careful about what the first one licenses. The adjacency certificate does not need causal sufficiency (Section 7), so the network's confounding does not by itself void its guarantee; but Remark 1's third requirement—a correctly specified per-set primitive—is not obviously met either, since we describe this same network as demonstrably nonlinear a paragraph below, and our default primitive is the linear-Gaussian safe test of Section 4.3. We have not verified that primitive's assumption on Sachs, so we cannot say the adjacency guarantee was live and satisfied here; what we can say is that all six certified edges agree with the consensus skeleton, which is empirical agreement, not a demonstration that the theorem's premises held. First, then: the experiment illustrates the asymmetry the paper is about—the adjacency certificate's six-for-six agreement is consistent with the guarantee it carries under weaker premises than orientation needs, while the certificate whose null requires a parametric functional model fails exactly where that model demonstrably fails, and does so without us having to guess at an unverified primitive assumption to explain it, since Section 5 states plainly what the direction null requires and this network does not have it. Second, the direction failure shows that the failure is not detectable from the output—the reversed arrowheads carry the same $\alpha = 0.1$ label as any other, which is precisely why Remark 6 names misspecification as the construction's main theoretical risk rather than a footnote. The conclusion we resist is that the consensus is wrong and the certificates are right; observational non-Gaussianity-based orientation disagreeing with interventional evidence is much more likely to be the model failing than the biology. Figure 11 shows how slowly the certified set accumulates.

Against a 2026 comparator on the same data.

Uehara (2026) evaluates the same observational Sachs recording ($n = 853$, 11 proteins) and reports, at a default operating point, direction precision 0.933 and recall 0.700 against 21 practitioner queries, with a cheaper 8-query configuration reaching precision and recall 1.000. Three differences make this a comparison of methods rather than a scoreboard. First, that protocol is *interactive*: it consumes expert answers to structural queries as part of reaching its numbers, where CERT-CD uses none—an interactive method that may ask a human is not doing the same task as one that may not, and a higher precision bought with 21 queries is not a free improvement over one bought with zero. Second, its reference network has 20 directed edges including a PKA$\leftrightarrow$PIP3 feedback pair, not the 17-arc acyclic consensus of Sachs et al. (2005) we certify against; a certificate-only method built on a DAG cannot certify a cycle, so the two evaluations are scored against different ground truths by construction, not by oversight. Third, recall is a well-posed quantity for a method that outputs a complete DAG at every operating point; for CERT-CD, at $\alpha = 0.1$ recall over the full 20- or 17-edge network is $6/17 \approx 0.35$ on the adjacency side and $0/17$ on the (wrong) orientation side, and neither number means what Uehara's recall means, since the undecided 49 pairs are not a rejection. We report the juxtaposition because it is the only case in this paper where a comparator we already cite for other reasons evaluates the identical real sample, not because the numbers settle anything against each other.

9.11 A real bivariate direction benchmark

Sachs is one dataset. A single real instance, however carefully analysed, cannot say whether Section 9.10's failure is characteristic of the direction certificate on real data or a property of that one confounded, nonlinear network. The standard corrective in the bivariate-direction literature is the Tübingen cause-effect pairs benchmark (Mooij, Peters, Janzing, Zscheischler, & Schölkopf, 2016): 108 real pairs drawn from 37 underlying datasets across meteorology, biology, medicine, engineering and economics, each with a direction the literature treats as ground truth—though, as the benchmark's own documentation states and we repeat rather than launder, that ground truth was not established by intervention. It is the benchmark LiNGAM, ANM, IGCI and RECI are conventionally scored against, and, as far as we can establish from the papers themselves, at least two of the 2025–2026 papers already cited in Section 8 for other reasons evaluate on it: Ding and Zhang (2025) and the statistical-power diagnostic of Prakash, Xia, and Erosheva (2026). Six of the 108 pairs have a multivariate cause or effect and are excluded, matching standard practice for this benchmark's bivariate accuracy figures; the excluded pair numbers are reported alongside the result rather than silently dropped.

Running the direction certificate on each of the remaining 102 pairs as an unconditioned bivariate test (Eqs. (7)–(8) with Z an intercept only), at $\alpha = 0.1$: it certifies a direction for 14 of 102 pairs, a coverage of 0.137. Coverage this low is itself informative—most of these pairs are close to linear only locally, or their noise is not well approximated by the generalised-Gaussian family of Eq. (11), and the certificate's response to a poor fit is to stay silent, exactly as Remark 6 says it should. What the low coverage does not buy is accuracy on the pairs where it does commit: of the 14 certified pairs, 6 agree with the benchmark's assumed ground truth, a weighted accuracy of 0.557, with a 95% Wilson interval of $[0.214, 0.674]$ that comfortably contains chance. The forced-choice comparison statistic—the same pairwise likelihood ratio DirectLiNGAM uses, run on every pair with no abstention available to it—reaches a weighted accuracy of 0.535. Mooij et al. (2016) report $63\% \pm 10\%$ accuracy (AUC

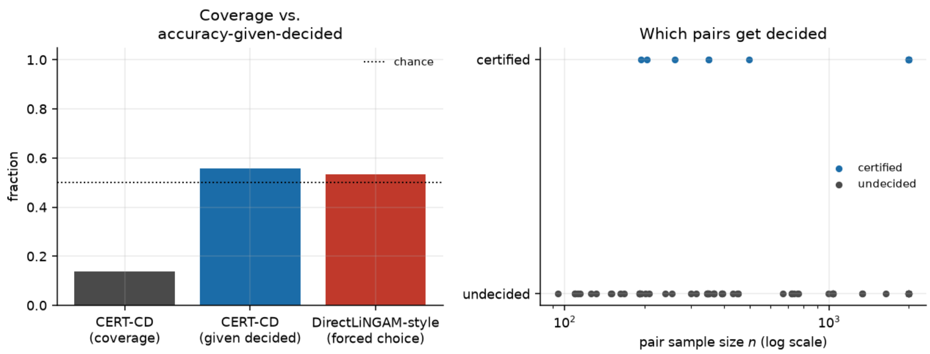

Fig. 10 The direction certificate on 102 Tübingen cause-effect pairs. Left: coverage (fraction of pairs certified), weighted accuracy conditional on certification, and the forced-choice comparison statistic's weighted accuracy, against chance. Right: which pairs get decided, by sample size.

0.74 ± 0.05) for the best-performing additive-noise-model method on this same real-world benchmark, and additive noise models are a strictly more general family than the linear one Eqs. (7)–(8) assume; a linear method landing well below that ceiling is the expected consequence of the benchmark's mechanisms being predominantly nonlinear, not a sign our implementation is behaving differently from other LiNGAM-family methods on the same data. Where the certificate does commit, it agrees with the forced-choice statistic on 12 of 14 pairs (0.857)—the same near-chance answer, reached the same way, on a benchmark whose mechanisms are known to be predominantly nonlinear where a strictly linear model like Eqs. (7)–(8) has no purchase. Figure 10 shows the coverage/accuracy split and which pairs, by sample size, get decided at all.

We take two things from this. First, low coverage under model misspecification is the certificate behaving as designed, and it is visible here at a scale Sachs alone could not demonstrate: thirteen percent commitment, not zero and not everything. Second, near-chance accuracy on the pairs it does commit to is a genuine limitation stated plainly rather than a result to defend: a certificate is a statement about a null model, and on a benchmark built to be hard for exactly that model, committing rarely does not guarantee committing correctly when the model happens to look locally plausible but is not. Both Sachs and Tübingen point at the same open problem named in Remark 6: bounding, rather than only measuring, what happens when the parametric family is wrong.

9.12 Where the certificates are void

The word "certificate" transfers authority, so the conditions under which a certificate means nothing have to be measured and named rather than left in a remark. We constructed four regimes, each breaking one premise of Theorem 9 while leaving the others intact, plus an in-model control, and ran 200 replications of each at $\alpha = 0.05$, $n = 3000$. Table 8 reports how often a false edge and a false arrowhead were certified.

The false-edge column is the same in every broken regime: a false edge is certified in every single run. This is worth stating without softening. The family-wise bound of Theorem 9 does not degrade gracefully when its premises fail; it stops applying, and

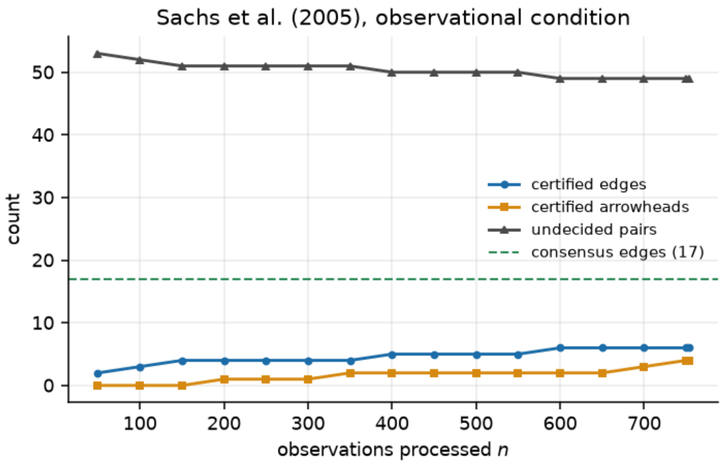

Fig. 11 CERT-CD on the Sachs observational sample. Certified edges accumulate slowly and the undecided set stays large: at $n = 753$, 49 of 55 pairs remain undecided.

Table 8 Probability that a false edge or a false arrowhead is ever certified, 200 replications per regime at $\alpha = 0.05$, $n = 3000$, Wilson 95% intervals. Each regime breaks exactly one premise. When a premise fails, the bound is *void*, not merely loose—and the table shows what void looks like.

regime	premise broken	false edge	false arrowhead
in-model control	none	0.000 [.000,.019]	0.000 [.000,.019]
separator needs $\lvert S\rvert = 3$, run at $k = 2$	Assumption 2	1.000 [.981,1.00]	0.130 [.090,.184]
latent common cause of the pair	causal sufficiency	1.000 [.981,1.00]	0.190 [.142,.250]
nonlinear mechanism	linearity	1.000 [.981,1.00]	0.710 [.644,.768]
Poisson counts, shared latent rate	linear-Gaussian model	1.000 [.981,1.00]	0.140 [.099,.195]

the observed error rate goes to one. A reader who takes $\alpha = 0.05$ from the output and does not check the premises is not getting a conservative answer, they are getting no guarantee at all.

Three of the four regimes break the *primitive* rather than the construction. PC with a Fisher-z test fails identically on the latent-confounder, nonlinear and count-data rows, for the same reason: a linear-Gaussian conditional-independence test cannot see a nonlinear dependence, and no causally sufficient method survives an unmeasured common cause. Proposition 3 is indifferent to the primitive (Remark 1), so substituting a nonparametric sequential test (He & Sutherland, 2026) would repair the last two rows at a cost in power. The first row is different: it breaks Assumption 2, which is a premise of the construction itself and cannot be repaired by changing the primitive, only by raising k.

The arrowhead column is the one that matters most in practice, because an arrowhead is what a practitioner acts on. It is lower than the edge column throughout—between 0.13 and 0.19 in three regimes, and 0.71 under a nonlinear mechanism—but "lower" is not "controlled". Every entry is far above the 0.025 that the direction half

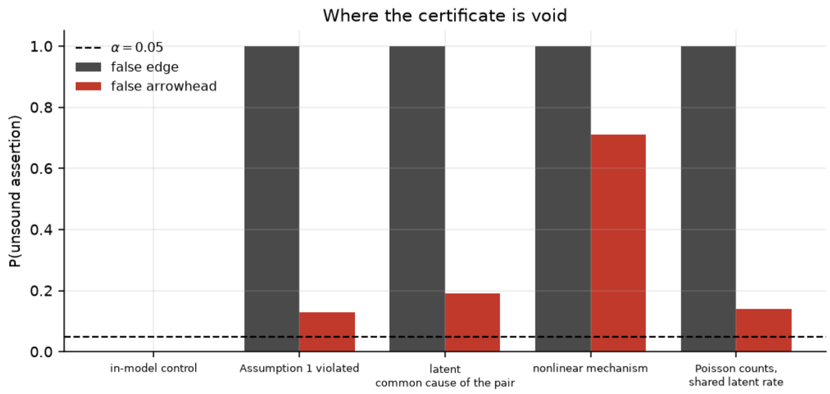

Fig. 12 Certified false edges and false arrowheads when one premise is broken at a time, against the in-model control. The dashed lines are the two halves of the budget. When a premise fails the bound does not degrade—it stops applying.

of the budget allows. Under a nonlinear mechanism the certificate is worse than uninformative: it points the wrong way in the majority of runs while carrying a label that says otherwise. This is the simulated counterpart of the protein-signalling result in Section 9.10, and the two should be read together. Figure 12 plots both columns against the two halves of the budget.

10 Discussion

10.1 What the output does not say

The absence of an edge in a CERT-CD graph means *undecided*. The algorithm cannot report that an edge is missing, and a pair may remain undecided indefinitely if adjacency-faithfulness fails for it—which is exactly the situation in which a delete-on-non-rejection method returns a confident wrong answer. This is the price of the rule. We regard stating it as preferable to concealing it, but it is a real cost: on the Sachs data the honest output is six certified edges and forty-nine shrugs, and a practitioner who needs a complete graph will not get one from this method. The optional fourth verdict of Section 6.5 recovers part of the missing information at the cost of naming a margin δ, and we think that trade is usually worth making in applications; we did not use it in the comparisons above because the comparators assert absence outright and the comparison should be with what they actually claim.

10.2 Limitations

We list these in what we take to be decreasing order of severity.

1. *The direction certificate's family-wise bound is not delivered when its own premise fails, and the premise is correlated with faithfulness.* Section 9.5 measures a wrong-arrow rate against budget on instances violating δ-strong faithfulness, and Section 9.5's block-completeness diagnostic traces the mechanism to the frozen

block more often omitting a true parent on those instances. Theorem 9 is not contradicted—its direction clause assumes the linear non-Gaussian model holds for the pair given its frozen block, and that assumption is what fails—but the practical reading of the theorem has to be adjusted: only the adjacency clause is robust to unfaithfulness. We do not have a fix to offer in its place. Freezing the conditioning block to all other variables, the obvious alternative, is not one: with no orientations available at freezing time it cannot exclude a collider or a descendant of either endpoint, and conditioning on either breaks the certifying-null property in a different way, even when every true parent is present. Finding a conditioning rule that avoids both failure modes is open.

2. *Misspecification of the direction certificate is not controlled more generally.* Proposition 7 requires the fitted supremum to dominate the likelihood at the true null parameter, which is guaranteed only when the noise lies in the fitted family. Section 9.10 shows this failing on real data, systematically and without any signal in the output. Bounding the gap—or replacing the generalised-Gaussian family with one whose supremum can be bounded under a nonparametric neighbourhood—is the most important open problem we leave.
3. *Assumption 2 can fail, and fails silently.* It is more often satisfied than δ-strong faithfulness (Table 3), but "more often" is not "always", and Section 9.12 shows that when it fails the family-wise bound is void rather than degraded. It is as unverifiable in practice as the assumption it replaces.
4. *Causal sufficiency for the direction certificate.* The arrowhead null assumes no unobserved common cause of the pair. Section 9.12 measures what latent confounding does; the answer is that arrowheads become unreliable well before adjacencies do. CERT-FCI (Section 7) drops the direction certificate for this reason, and its FCI-rule orientations carry no family-wise guarantee.
5. *The multiplicity layer is a union bound.* Section 6.4 explains why nothing sharper is currently available for dependent pseudo e-values, and that a solution would improve every result here.
6. *Scaling is unproven.* The direction layer refits per pair per batch and does not reduce to a sufficient statistic; experiments reach $d = 11$ on real data and $d = 7$ in simulation. Section 6.3 identifies two straightforward optimisations we have not implemented, but we make no claim about behaviour at d in the hundreds.
7. *The FCI variant is incomplete.* It implements orientation rules R1–R4 and omits R5–R10, so its PAG is sound but not maximally informative.
8. *Power is unquantified in theory.* We report certification rates empirically throughout, but we prove nothing about them. A power analysis under adjacency-faithfulness with a stated margin would be a natural companion result and we do not have one.

10.3 Outlook

Three directions seem worth pursuing. The first is the multiplicity problem of Section 6.4: family-wise control for dependent pseudo e-values would be useful well beyond causal discovery. The second is replacing the generalised-Gaussian direction likelihoods with a family admitting a computable supremum under misspecification, which would close the gap that Section 9.10 exposes; the reverse-regression characterisation of Ding and Zhang (2025) and the nonparametric route of He and Sutherland (2026) both look promising. The third is interventional data, where the certificate-only rule has an obvious analogue—an intervention that changes a variable's distribution certifies an ancestral relation by rejecting invariance—and where Peters et al. (2016) already provides the one-sided logic that Section 9.9 makes anytime-valid.

11 Conclusion

Constraint-based causal discovery deletes edges on non-rejection, and this single design decision is the source of both its dependence on faithfulness for finite-sample correctness and its inability to distinguish an established absence from an unexamined pair. We asked what an algorithm looks like if it refuses to do that: assert only by rejecting a null that is true whenever the asserted feature is absent.

The answer is an algorithm that grows from the empty graph and reports undecided pairs as undecided. Adjacency is certified by an e-process for a union null requiring the Markov condition and a separator-size bound but no faithfulness, and the resulting bound—the probability that any certified edge is ever absent, uniformly over time and over the whole graph—holds unconditionally. Orientation is certified by sequential universal inference against the reversed factorisation, escaping the Markov equivalence class and abstaining automatically under Gaussianity; the corresponding bound on a wrong arrowhead holds only conditional on the linear non-Gaussian model actually describing the pair given its frozen conditioning block, and Section 9.5 measures that condition failing on instances violating δ-strong faithfulness, where the block is more often incomplete. What the design costs beyond that conditionality is power, and we have tried to report both as prominently as the guarantee—in the power curves of Section 9.2, the undecided fractions of Section 9.5, and the six-edges-out-of-fifty-five of Section 9.10.

The real-data experiments deliver the paper's own counterexamples, and we are careful about which side is which. On Sachs the six certified edges agree with the consensus skeleton, which is empirical agreement, not a demonstration that the adjacency certificate's guarantee was live: we did not verify the linear-Gaussian primitive on data we ourselves call demonstrably nonlinear, so we do not claim the theorem's premises held there, only that the output happened to be right. Where the certifying null instead rests on a parametric functional model, the failure is visible and unambiguous: on Sachs the direction certificate is confidently and systematically wrong, with nothing in its output to say so; on the Tübingen pairs it mostly declines to commit, correctly, but is no better than chance on the minority of pairs where it does. A certificate is only ever as good as the null it certifies against, and we think making that dependence visible—rather than absorbing it into an assumption stated once and not revisited, or into a success we cannot actually attribute to a live guarantee—is the more useful contribution.

Statements and Declarations

Competing interests.

The author declares no competing interests.

Funding.

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Ethics approval.

Not applicable. The study involves no human participants or animals. The protein-signalling data analysed in Section 9.10 are publicly available and were published by Sachs et al. (2005).

Data and code availability.

All simulation code, experiment scripts, random seeds and figure-generating scripts are available at https://github.com/vinhqdang/SPRINT-CD. The protein-signalling data of Sachs et al. (2005) are redistributed with the `bnlearn` package (Scutari, 2010) and are downloaded by the experiment script rather than included in the repository. The Tübingen cause-effect pairs of Section 9.11 are likewise downloaded on first use from https://webdav.tuebingen.mpg.de/cause-effect/ and are not redistributed.

Author contributions.

The author is solely responsible for all aspects of this work.

Use of generative AI in the preparation of this manuscript.

A large language model was used in support of writing this manuscript. The author verified and modified the resulting content and takes full responsibility for the manuscript in its entirety, including its claims, derivations, experiments and references.

Appendix A The safe linear e-process

We derive (5) and prove Proposition 4.

A.1 Setup

Write $y \in \mathbb{R}^n$ for the stacked observations of X_i, $x \in \mathbb{R}^n$ for those of X_j, and $W \in \mathbb{R}^{n\times q}$ for the design matrix of the conditioning block X_S together with an intercept, $q = |S| + 1$. The model is

$$y \;=\; W\alpha \;+\; \beta x \;+\; \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2 I_n), \tag{A1}$$

with $H_0 : \beta = 0$ and $H_1 : \beta \neq 0$, and nuisance parameters $(\alpha, \sigma) \in \mathbb{R}^q \times (0, \infty)$ common to both.

Let $M = I_n - W(W^\top W)^{-1}W^\top$ be the residual projector onto the orthogonal complement of the column space of W, and write $\tilde{y} = My$, $\tilde{x} = Mx$ for the residualised vectors. Set

$$S_{yy} = \tilde{x}^\top \tilde{x}, \qquad S_{xy} = \tilde{x}^\top \tilde{y}, \qquad S_{xx} = \tilde{y}^\top \tilde{y},$$

following the naming convention of the implementation, in which the first subscript refers to the *response* of the regression being examined. The squared sample partial correlation is $\hat{r}^2 = S_{xy}^2/(S_{xx}S_{yy})$, and the least-squares estimate of β is $\hat{\beta} = S_{xy}/S_{yy}$.

A.2 Invariance and the right-Haar prior

The null model (A1) with $\beta = 0$ is invariant under the group G of transformations

$$g_{c,\lambda}: \quad y \mapsto \lambda y + Wc, \qquad c \in \mathbb{R}^q,\ \lambda > 0,$$

acting on the parameters by $\alpha \mapsto \lambda\alpha + c$, $\sigma \mapsto \lambda\sigma$, and—under the alternative—$\beta \mapsto \lambda\beta$ once x is transformed correspondingly. The right-Haar measure for this group is $\pi(\alpha, \sigma)\, d\alpha\, d\sigma \propto \sigma^{-1}\, d\alpha\, d\sigma$.

The relevant fact (Berger et al., 1998; Hendriksen et al., 2021) is that when the same right-Haar prior is placed on a nuisance group shared by both hypotheses, the resulting Bayes factor equals the Bayes factor based on the maximal invariant, whose distribution under the null does not depend on the nuisance parameters. That equality gives a martingale property in the filtration generated by the maximal invariants at each n; extending it to the (finer) filtration generated by the raw observations—the one Proposition 4 and Definition 2 use—needs G to be *amenable*, so that the right-Haar measure can be approximated by a sequence of proper, group-invariant priors and the resulting Bayes factor remains a martingale in the unreduced filtration (Hendriksen et al., 2021). G here is the semidirect product of the translations $\mathbb{R}^q$ (abelian) and the positive scalings $\mathbb{R}_{>0}$ (also abelian), a solvable Lie group, and every solvable group is amenable; this is the standard route by which the right-Haar construction is applied sequentially rather than only at a fixed n, and we make the appeal to it explicit here rather than leaving it implicit in the citation. Consequently the Bayes factor is a *test martingale* in the observation filtration: its expectation under any null distribution, conditional on the past, is one, whatever the nuisance parameters happen to be. This is the property that fails for a generic Bayes factor and that makes optional stopping legitimate here.

A.3 Integrating the marginal likelihoods

Under H_0, integrating (A1) against $\pi(\alpha, \sigma) \propto \sigma^{-1}$ gives

$$m_0(y) \;=\; \int_0^\infty \int_{\mathbb{R}^q} (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\|y - W\alpha\|^2}{2\sigma^2}\right\} \frac{d\alpha\, d\sigma}{\sigma} \;=\; C_{n,q}\, \left(S_{xx}\right)^{-(n-q)/2}, \tag{A2}$$

where $C_{n,q} = \frac{1}{2}\pi^{-(n-q)/2}\Gamma\left(\frac{n-q}{2}\right)|W^\top W|^{-1/2}$ collects the terms that do not depend on the data through the residuals. The α-integral is Gaussian and contributes $|W^\top W|^{-1/2}$

together with the replacement of $\|y - W\alpha\|^2$ by $\|\tilde{y}\|^2 = S_{xx}$; the σ-integral is then a gamma integral.

Under H_1 we add $\beta \mid \sigma \sim N(0, \sigma^2 g)$, independent of α. Completing the square in β against the residualised design gives a Gaussian integral with precision $\sigma^{-2}(S_{yy}+g^{-1})$ and mean $S_{xy}/(S_{yy}+g^{-1})$, so

$$\begin{aligned} m_1(y) &= C_{n,q}\left(1 + gS_{yy}\right)^{-1/2}\left(S_{xx} - \frac{S_{xy}^2}{S_{yy}+g^{-1}}\right)^{-(n-q)/2} \\ &= C_{n,q}\,(1+G)^{-1/2}\, S_{xx}^{-(n-q)/2}\left(1 - \kappa\hat{r}^2\right)^{-(n-q)/2}, \end{aligned} \tag{A3}$$

where $G = gS_{yy}$ and $\kappa = G/(1+G)$, using $\dfrac{S_{xy}^2}{S_{xx}(S_{yy}+g^{-1})} = \dfrac{S_{xy}^2}{S_{xx}S_{yy}} \cdot \dfrac{S_{yy}}{S_{yy}+g^{-1}} = \hat{r}^2\kappa$.

Dividing (A3) by (A2), the constants and the $S_{xx}^{-(n-q)/2}$ factors cancel:

$$E_n \;=\; \frac{m_1(y)}{m_0(y)} \;=\; (1+G)^{-1/2}\left(1 - \kappa\hat{r}_n^2\right)^{-(n-q)/2}, \tag{A4}$$

which is (5) with $p = q = |S| + 1$. Proposition 4 follows from the invariance argument above: (E_n) is a ratio of marginal likelihoods under a shared right-Haar prior, so it is a test martingale under every null distribution, and by optional stopping an e-process.

A.4 Two implementation notes

Fixing g.

$G = gS_{yy}$ grows with n because S_{yy} does. The martingale property requires g to be $\mathcal{F}_0$-measurable; setting it from the running data—for instance to hold G constant—breaks it. We choose g from a warm-up prefix that is discarded, so it is a constant as far as every e-process is concerned.

Streaming.

All of S_{xx}, S_{xy}, S_{yy} are Schur complements of the running Gram matrix of the full observation vector. Maintaining that matrix costs $O(d^2)$ per observation and serves every (i, j, S) triple, so adding hypotheses costs computation at query time but no additional state.

Appendix B Inverting the e-process into a confidence sequence

Fix β_0 and consider the null $\beta = \beta_0$. Replacing y by $y - \beta_0 x$ in (A4) gives the e-process for that null; the confidence sequence at level $1 - \alpha$ is the set of β_0 not rejected,

$$\left\{\beta_0 : (1+G)^{-1/2}\left(1 - \kappa\hat{r}^2(\beta_0)\right)^{-(n-q)/2} < 1/\alpha\right\},$$

where $\hat{r}^2(\beta_0)$ is the squared partial correlation computed from $y - \beta_0 x$. Taking logarithms, the condition reads

$$-\tfrac{1}{2}\log(1+G) - \tfrac{n-q}{2}\log\bigl(1-\kappa\hat{r}^2(\beta_0)\bigr) < -\log\alpha,$$

that is $\hat{r}^2(\beta_0) < \tau$ with

$$\tau \;=\; \frac{1}{\kappa}\left[1-\exp\!\left(\frac{2}{n-q}\Bigl(\log\alpha-\tfrac{1}{2}\log(1+G)\Bigr)\right)\right] \;=\; \frac{1}{\kappa}\left[1-\left(\frac{\alpha}{\sqrt{1+G}}\right)^{2/(n-q)}\right]. \tag{B5}$$

Writing $S_{xy}(\beta_0) = S_{xy} - \beta_0 S_{yy}$ and $S_{xx}(\beta_0) = S_{xx} - 2\beta_0 S_{xy} + \beta_0^2 S_{yy}$, the condition $S_{xy}(\beta_0)^2 < \tau\, S_{xx}(\beta_0) S_{yy}$ is a quadratic inequality in β_0 with leading coefficient $(1-\tau)S_{yy}^2 > 0$ whenever $\tau < 1$, and solving it gives

$$\hat{\beta} \;\pm\; \frac{1}{S_{yy}}\sqrt{\frac{\tau\bigl(S_{xx}S_{yy}-S_{xy}^2\bigr)}{1-\tau}}, \qquad \hat{\beta} = S_{xy}/S_{yy}, \tag{B6}$$

which is (6). When $\tau \le 0$—which happens at small n, since $(\alpha/\sqrt{1+G})^{2/(n-q)} \to 1$ from above as $n \downarrow q$—the inequality has no solution and the interval is the whole line.

Remark 8 (A sign that mattered) The bracketed term in (B5) carries $-\frac{1}{2}\log(1+G)$, not $+\frac{1}{2}\log(1+G)$. With the sign reversed, the formula still produces intervals of a plausible shape, still shrinks at the right rate, and still passes every Type-I error simulation of the underlying *test*, because the test is unaffected. It produces intervals roughly half the correct width. We caught it only by comparing (B6) against a brute-force numerical inversion of (A4) over a grid of β_0, which now runs as a unit test. Closed-form inversions of e-processes deserve this check as a matter of routine.

Appendix C The frozen conditioning block

Proposition 7 is stated for a fixed null. In Algorithm 1 the conditioning block Z_{ij} is selected at the random time T at which the pair's adjacency is certified, so the hypothesis being tested is itself chosen using data. We record why the guarantee survives.

Let T be the certification time of pair (i,j); it is a stopping time with respect to $(\mathcal{F}_t)$, since certification is triggered by the running maximum of $A_t(i,j)$, which is adapted. Let Z_{ij} be $\mathcal{F}_T$-measurable (it is: it is a function of the certified statuses at time T). The direction process is built only from observations $o_{T+1}, o_{T+2}, \ldots$.

Condition on $\mathcal{F}_T$. Because the observations are i.i.d. and T is a stopping time, the post-T observations are, conditionally on $\mathcal{F}_T$, i.i.d. from P; and Z_{ij} is now a fixed set. The proof of Proposition 7 therefore applies verbatim to the conditional law, giving $\mathbb{E}_P[E_{T+\tau} \mid \mathcal{F}_T] \le 1$ for every stopping time τ of the post-T filtration, on the event $\{T < \infty\}$. Taking expectations, $\mathbb{E}_P[E_{T+\tau}\mathbf{1}\{T<\infty\}] \le \mathbb{P}(T<\infty) \le 1$, and Ville's inequality applies to the post-T process. Defining $E_t = 1$ for $t \leq T$ extends this to a process on the original time index without affecting the bound.

Two remarks. First, this is why the implementation must discard all observations before T: reusing them would break the conditional independence, and the constant they contribute is exactly the failure mode of Remark 5. Second, the argument is what licenses the graph-dependent execution order in Theorem 9: the algorithm may decide *which* direction hypotheses to examine using the data, because the union bound runs over the full fixed family $\{(i, j)\}$ regardless, and each member of that family has a valid e-process whether or not the algorithm chooses to evaluate it.

Appendix D Additional experimental detail

Table D1 Realised crossing rates of the per-set primitive under six noise laws, 2000 replications each, monitored to $n = 800$ after every tenth observation. Every entry is at or below its nominal level; the largest realised-to-nominal ratio in the table is 0.95 (shifted exponential, $\alpha = 0.01$) and the median is 0.49.

noise law	$\alpha = 0.20$	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
Gaussian	0.091	0.046	0.023	0.003
Laplace	0.092	0.047	0.024	0.003
uniform	0.107	0.052	0.031	0.006
Student-t_5	0.093	0.044	0.020	0.004
shifted exponential	0.101	0.051	0.028	0.010
two-component mixture	0.097	0.049	0.028	0.006

Hyperparameters.

Unless stated otherwise: warm-up $m = 50$ observations (discarded), prior standard deviation 0.5 after standardisation—so $g = 0.25$ in the parameterisation of Eq. (5), and $g = 0.16$ in the primitive-calibration experiment, which uses 0.4— $k = 2$, adjacency share $\alpha_A/\alpha = 0.5$, batch size 250 in the graph experiments and 100 in the skeleton experiments, and a minimum of 80 post-certification observations before a direction process is first evaluated. The equivalence-region width for SPRINT-CD is $\delta = 0.15$ on the standardised scale, matching the δ used to define the strata.

Reproduction.

`python experiments/run_all.py` reproduces every figure and table; individual scripts accept `--quick` for a low-replication smoke test, which writes to `results/quick/` so that it cannot overwrite the full-precision artefacts the paper cites. Seeds are fixed in each script.

References

Amin, A.N., & Wilson, A.G. (2024). Scalable and flexible causal discovery with an efficient test for adjacency. *Proceedings of the 41st international conference on machine learning* (Vol. 235).

Benjamini, Y., & Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, *29*(4), 1165–1188, https://doi.org/10.1214/aos/1013699998

Berger, J.O., Pericchi, L.R., Varshavsky, J.A. (1998). Bayes factors and marginal distributions in invariant situations. *Sankhyā: The Indian Journal of Statistics Series A*, *60*(3), 307–321,

Box, G.E.P., & Tiao, G.C. (1973). *Bayesian inference in statistical analysis*. Reading, MA: Addison-Wesley.

Bühlmann, P., Peters, J., Ernest, J. (2014). CAM: Causal additive models, high-dimensional order search and penalized regression. *The Annals of Statistics*, *42*(6), 2526–2556, https://doi.org/10.1214/14-AOS1260

Candès, E., Fan, Y., Janson, L., Lv, J. (2018). Panning for gold: Model-X knockoffs for high-dimensional controlled variable selection. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *80*(3), 551–577, https://doi.org/10.1111/rssb.12265

Chickering, D.M. (2002). Optimal structure identification with greedy search. *Journal of Machine Learning Research*, *3*, 507–554,

Colombo, D., Maathuis, M.H., Kalisch, M., Richardson, T.S. (2012). Learning high-dimensional directed acyclic graphs with latent and selection variables. *The Annals of Statistics*, *40*(1), 294–321, https://doi.org/10.1214/11-AOS940

Comon, P. (1994). Independent component analysis, a new concept? *Signal Processing*, *36*(3), 287–314, https://doi.org/10.1016/0165-1684(94)90029-9

Csillag, D., Struchiner, C.J., Goedert, G.T. (2025). Prediction-powered e-values. *Proceedings of the 42nd international conference on machine learning.* (arXiv:2502.04294)

Darmois, G. (1953). Analyse générale des liaisons stochastiques. *Revue de l'Institut International de Statistique*, *21*(1/2), 2–8, https://doi.org/10.2307/1401511

Ding, Z., & Zhang, X.-P. (2025). *GaussDetect-LiNGAM: Causal direction identification without Gaussianity test.* (Preprint, arXiv:2512.03428)

Grünwald, P., de Heide, R., Koolen, W.M. (2024). Safe testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *86*(5), 1091–1128, https://doi.org/10.1093/jrsssb/qkae011

Hartog, W., & Lei, L. (2026). *Family-wise error rate control with e-values.* (Preprint, arXiv:2501.09015)

He, Z., & Sutherland, D.J. (2026). *Sequential kernel-based conditional independence testing via adaptive betting.* (Preprint, arXiv:2606.18993)

Hendriksen, A., de Heide, R., Grünwald, P. (2021). Optional stopping with Bayes factors: A categorization and extension of folklore results, with an application to invariant situations. *Bayesian Analysis*, *16*(3), 961–989, https://doi.org/10.1214/20-BA1234

Howard, S.R., Ramdas, A., McAuliffe, J., Sekhon, J. (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, *49*(2), 1055–1080, https://doi.org/10.1214/20-AOS1991

Hoyer, P.O., Janzing, D., Mooij, J.M., Peters, J., Schölkopf, B. (2009). Nonlinear causal discovery with additive noise models. *Advances in neural information processing systems 21* (pp. 689–696).

Hyvärinen, A. (1998). New approximations of differential entropy for independent component analysis and projection pursuit. *Advances in Neural Information Processing Systems*, *10*, 273–279,

Hyvärinen, A., & Smith, S.M. (2013). Pairwise likelihood ratios for estimation of non-Gaussian structural equation models. *Journal of Machine Learning Research*, *14*, 111–152,

Jeffreys, H. (1946). An invariant form for the prior probability in estimation problems. *Proceedings of the Royal Society of London A*, *186*(1007), 453–461, https://doi.org/10.1098/rspa.1946.0056

Johari, R., Koomen, P., Pekelis, L., Walsh, D. (2022). Always valid inference: Continuous monitoring of A/B tests. *Operations Research*, *70*(3), 1806–1821, https://doi.org/10.1287/opre.2021.2135

Kalisch, M., & Bühlmann, P. (2007). Estimating high-dimensional directed acyclic graphs with the PC-algorithm. *Journal of Machine Learning Research*, *8*, 613–636,

Komatsu, Y., Shimizu, S., Shimodaira, H. (2010). Assessing statistical reliability of LiNGAM via multiscale bootstrap. *Artificial neural networks – icann 2010* (Vol. 6354, pp. 309–314). Berlin: Springer.

Lindon, M., Ham, D.W., Tingley, M., Bojinov, I. (2026). Anytime-valid inference in linear models with applications to regression-adjusted causal inference. *Journal of the American Statistical Association*, 1–12, https://doi.org/10.1080/01621459.2026.2692052

Maathuis, M.H., Kalisch, M., Bühlmann, P. (2009). Estimating high-dimensional intervention effects from observational data. *The Annals of Statistics*, *37*(6A), 3133–3164, https://doi.org/10.1214/09-AOS685

Meek, C. (1995a). Causal inference and causal explanation with background knowledge. *Proceedings of the 11th conference on uncertainty in artificial intelligence* (pp. 403–410).

Meek, C. (1995b). Strong completeness and faithfulness in Bayesian networks. *Proceedings of the 11th conference on uncertainty in artificial intelligence* (pp. 411–418).

Mooij, J.M., Peters, J., Janzing, D., Zscheischler, J., Schölkopf, B. (2016). Distinguishing cause from effect using observational data: Methods and benchmarks. *Journal of Machine Learning Research*, *17*(32), 1–102,

Pearl, J. (2009). Causal inference in statistics: An overview. *Statistics Surveys*, *3*, 96–146, https://doi.org/10.1214/09-SS057

Peters, J., & Bühlmann, P. (2014). Identifiability of Gaussian structural equation models with equal error variances. *Biometrika*, *101*(1), 219–228, https://doi.org/10.1093/biomet/ast043

Peters, J., Bühlmann, P., Meinshausen, N. (2016). Causal inference by using invariant prediction: Identification and confidence intervals. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *78*(5), 947–1012, https://doi.org/10.1111/rssb.12167

Peters, J., Janzing, D., Schölkopf, B. (2017). *Elements of causal inference: Foundations and learning algorithms.* Cambridge, MA: MIT Press.

Prakash, S., Xia, F., Erosheva, E. (2024). *A diagnostic tool for functional causal discovery.* (Preprint, arXiv:2406.07787)

Prakash, S., Xia, F., Erosheva, E.A. (2026). *Causal discovery via statistical power.* (Preprint, arXiv:2605.13550)

Ramdas, A., Grünwald, P., Vovk, V., Shafer, G. (2023). Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, *38*(4), 576–601, https://doi.org/10.1214/23-STS894

Ramsey, J., Spirtes, P., Zhang, J. (2006). Adjacency-faithfulness and conservative causal inference. *Proceedings of the 22nd conference on uncertainty in artificial intelligence* (pp. 401–408).

Richardson, T., & Spirtes, P. (2002). Ancestral graph Markov models. *The Annals of Statistics*, *30*(4), 962–1030, https://doi.org/10.1214/aos/1031689015

Robins, J.M., Scheines, R., Spirtes, P., Wasserman, L. (2003). Uniform consistency in causal inference. *Biometrika*, *90*(3), 491–515, https://doi.org/10.1093/biomet/90.3.491

Ruiz, G., Madrid Padilla, O.H., Zhou, Q. (2022). *Sequentially learning the topological ordering of causal directed acyclic graphs with likelihood ratio scores.* (Preprint, arXiv:2202.01748)

Sachs, K., Perez, O., Pe'er, D., Lauffenburger, D.A., Nolan, G.P. (2005). Causal protein-signaling networks derived from multiparameter single-cell data. *Science*, *308*(5721), 523–529, https://doi.org/10.1126/science.1105809

Scutari, M. (2010). Learning Bayesian networks with the bnlearn R package. *Journal of Statistical Software*, *35*(3), 1–22, https://doi.org/10.18637/jss.v035.i03

Shafer, G., & Vovk, V. (2019). *Game-theoretic foundations for probability and finance.* Hoboken, NJ: Wiley.

Shah, R.D., & Peters, J. (2020). The hardness of conditional independence testing and the generalised covariance measure. *The Annals of Statistics*, *48*(3), 1514–1538, https://doi.org/10.1214/19-AOS1857

Shaska, J., & Mitra, U. (2025). *Causal link discovery with unequal edge error tolerance.* (Preprint, arXiv:2507.21570)

Shimizu, S., Hoyer, P.O., Hyvärinen, A., Kerminen, A. (2006). A linear non-Gaussian acyclic model for causal discovery. *Journal of Machine Learning Research*, *7*, 2003–2030,

Shimizu, S., Inazumi, T., Sogawa, Y., Hyvärinen, A., Kawahara, Y., Washio, T., ... Bollen, K. (2011). DirectLiNGAM: A direct method for learning a linear non-Gaussian structural equation model. *Journal of Machine Learning Research*, *12*, 1225–1248,

Shimizu, S., & Kano, Y. (2008). Use of non-normality in structural equation modeling: Application to direction of causation. *Journal of Statistical Planning and Inference*, *138*(11), 3483–3491, https://doi.org/10.1016/j.jspi.2006.01.017

Solus, L., Wang, Y., Uhler, C. (2021). Consistency guarantees for greedy permutation-based causal inference algorithms. *Biometrika*, *108*(4), 795–814, https://doi.org/10.1093/biomet/asaa104

Spirtes, P. (2001). An anytime algorithm for causal inference. *Proceedings of the 8th international workshop on artificial intelligence and statistics* (pp. 213–221).

Spirtes, P., Glymour, C., Scheines, R. (2000). *Causation, prediction, and search* (2nd ed.). Cambridge, MA: MIT Press.

Spirtes, P., Meek, C., Richardson, T. (1995). Causal inference in the presence of latent variables and selection bias. *Proceedings of the 11th conference on uncertainty in artificial intelligence* (pp. 499–506).

Strieder, D., & Drton, M. (2023). Confidence in causal inference under structure uncertainty in linear causal models with equal variances. *Journal of Causal Inference*, *11*(1), 20230030, https://doi.org/10.1515/jci-2023-0030

Strobl, E.V., Spirtes, P.L., Visweswaran, S. (2019). Estimating and controlling the false discovery rate of the PC algorithm using edge-specific p-values. *ACM Transactions on Intelligent Systems and Technology*, *10*(5), 1–37, https://doi.org/10.1145/3351342

Subbotin, M.T. (1923). On the law of frequency of error. *Matematicheskii Sbornik*, *31*(2), 296–301,

Tse, T., & Davison, A.C. (2022). A note on universal inference. *Stat*, *11*(1), e501, https://doi.org/10.1002/sta4.501

Turner, R.J., & Grünwald, P.D. (2023). Safe sequential testing and effect estimation in stratified count data. *Proceedings of the 26th international conference on artificial intelligence and statistics* (Vol. 206, pp. 4880–4893).

Uehara, E. (2026). *Iterative causal discovery: Per-edge impossibility certificates, tier-aware oracle queries, and the* $1{+}k$ *lower bound.* (Preprint, arXiv:2605.27477)

Uhler, C., Raskutti, G., Bühlmann, P., Yu, B. (2013). Geometry of the faithfulness assumption in causal inference. *The Annals of Statistics*, *41*(2), 436–463, https://doi.org/10.1214/12-AOS1080

Ville, J. (1939). *Étude critique de la notion de collectif.* Paris: Gauthier-Villars.

Vovk, V., & Wang, R. (2021). E-values: Calibration, combination and applications. *The Annals of Statistics*, *49*(3), 1736–1754, https://doi.org/10.1214/20-AOS2020

Vovk, V., & Wang, R. (2024). Merging sequential e-values via martingales. *Electronic Journal of Statistics*, *18*(1), 1185–1205, https://doi.org/10.1214/24-EJS2228

Wang, R., & Ramdas, A. (2022). False discovery rate control with e-values. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *84*(3), 822–852, https://doi.org/10.1111/rssb.12489

Wasserman, L., Ramdas, A., Balakrishnan, S. (2020). Universal inference. *Proceedings of the National Academy of Sciences*, *117*(29), 16880–16890, https://doi.org/10.1073/pnas.1922664117

Waudby-Smith, I., & Ramdas, A. (2024). Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *86*(1), 1–27, https://doi.org/10.1093/jrsssb/qkad009

Zellner, A. (1986). On assessing prior distributions and Bayesian regression analysis with g-prior distributions. P.K. Goel & A. Zellner (Eds.), *Bayesian inference and decision techniques* (pp. 233–243). Amsterdam: North-Holland.

Zhang, J. (2008). On the completeness of orientation rules for causal discovery in the presence of latent confounders and selection bias. *Artificial Intelligence*, *172*(16-17), 1873–1896, https://doi.org/10.1016/j.artint.2008.08.001

Zhang, J., & Spirtes, P. (2008). Detection of unfaithfulness and robust causal inference. *Minds and Machines*, *18*(2), 239–271, https://doi.org/10.1007/s11023-008-9096-4

Zhang, K., & Hyvärinen, A. (2009). On the identifiability of the post-nonlinear causal model. *Proceedings of the 25th conference on uncertainty in artificial intelligence* (pp. 647–655).